

1. BINOMIAL THEOREM

SYNOPSIS

Binomial theorem for positive integral index and some key factors related to Binomial theorem

1. If n is a positive integer then

$$(x+a)^n = \sum_{r=0}^n {}^nC_r x^{n-r} a^r = {}^nC_0 x^n + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + \dots + {}^nC_n a^n$$

- The number of terms in the expansion $(x+a)^n$ is $n+1$
- The sum of the powers of x and a in any term in the expansion of $(x+a)^n$ is n
- The general term in the expansion of $(x+a)^n$ is $T_{r+1} = {}^nC_r x^{n-r} a^r$
- ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are binomial coefficients in the expansion of $(x+a)^n$
- The binomial coefficients which are equidistant from the beginning and from the ending are equal.
i.e. ${}^nC_0 = {}^nC_n, {}^nC_1 = {}^nC_{n-1}, {}^nC_2 = {}^nC_{n-2}, \dots$ etc.
- In the expansion of $(x+a)^n$, r^{th} term from the end is equal to $(n-r+2)^{\text{th}}$ term from the beginning.

2. $(x-a)^n = \sum_{r=0}^n (-1)^r {}^nC_r x^{n-r} a^r = {}^nC_0 x^n - {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 - \dots + (-1)^n {}^nC_n a^n$

The general term in this expansion is $T_{r+1} = (-1)^r {}^nC_r x^{n-r} a^r$

Note : The expansions of $(x+a)^n$ and $(a+x)^n$ are equal but their respective terms are not equal.

3. $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$

The general term in the expansion $T_{r+1} = {}^nC_r x^r$

4. $(1-x)^n = \sum_{r=0}^n (-1)^r {}^nC_r x^r = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - \dots + (-1)^n {}^nC_n x^n$

The general term in the expansion $T_{r+1} = (-1)^r {}^nC_r x^r$

Number of terms

5. a) The number of non-zero terms in the expansion of $\{(x+a)^n + (x-a)^n\}$ is

i) $\frac{n+1}{2}$, if n is an odd integer. ii) $\frac{n}{2} + 1$, if n is even integer.

- b) The number of non-zero terms in the expansion of $\{(x+a)^n - (x-a)^n\}$ is

i) $\frac{n+1}{2}$, if n is an odd integer. ii) $\frac{n}{2}$, if n is even integer.

- c) The number of terms in the expansion of $(x+a)^n + (x-a)^n + (x+ai)^n$ is $\left\lceil \frac{n+4}{4} \right\rceil$. When $\lceil . \rceil$ is G.I.F.

6. i) If n is **odd** there will be **two middle** terms in the expansion $(x+a)^n$ which are $\left(\frac{n+1}{2}\right)^{th}$ and $\left(\frac{n+3}{2}\right)^{th}$ terms
- ii) If n is **even** there will be only **one middle** term in the expansion $(x+a)^n$, which is $\left(\frac{n}{2}+1\right)^{th}$ term
7. The coefficient of the middle term is the greatest binomial coefficient in the expansion of $(x+a)^n$
8. i) If n is **odd**, there are **two** greatest binomial co-efficients in the expansion which are ${}^nC_{\frac{n-1}{2}}$ and ${}^nC_{\frac{n+1}{2}}$ also ${}^nC_{n-1/2} = {}^nC_{n+1/2}$.
- ii) If n is **even**, there is only **one** greatest binomial coefficient in the expansion $(x+a)^n$ which is ${}^nC_{n/2}$
9. The coefficient of x^k in the expansion of $\left(ax^p + \frac{b}{x^q}\right)^n$ is ${}^nC_r a^{n-r} b^r$ where $r = \frac{np-k}{p+q}$
10. The term independent of x in the expansion of $\left(ax^p + \frac{b}{x^q}\right)^n$ is ${}^nC_r a^{n-r} b^r$ where $r = \frac{np}{p+q}$
11. a) A generalised multinomial theorem

$$(x_1 + x_2 + x_3 + \dots + x_m)^n = \sum \frac{n!}{n_1! n_2! \dots n_m!} x_1^{n_1} x_2^{n_2} \dots x_m^{n_m}$$

Where the summation is taken over all non negative integers $n_1, n_2, \dots, n_m$ such that $n_1 + n_2 + \dots + n_m = n$

b) The general term in the expansion of $(x_1 + x_2 + \dots + x_m)^n$ is $\frac{(n_1 + n_2 + \dots + x_m^{n_m})!}{n_1! n_2! \dots n_m!} (x_1^{n_1} x_2^{n_2} \dots x_m^{n_m})$

c) No. of terms in the expansion of $(x_1 + x_2 + \dots + x_m)^n$ is ${}^{(n+m-1)}C_{(m-1)}$

d) The number of terms in the expansion of $(x + y + z)^n$ is $\frac{(n+1)(n+2)}{2}$.

e) The greatest coefficient in the expansion of $(x_1 + x_2 + \dots + x_m)^n$ is equal to $\frac{n!}{(q!)^{m-r} ((q+1)!)^r}$

where q is the quotient and r is the remainder when n is divided by m

12. **Numerically greatest term (N.G.T.)** in the expansion of $(1+x)^n$

i) If $\frac{(n+1)|x|}{|x|+1} = P + f$, then there exists only **one** N.G.T. which is $(P+1)^{th}$ term and its value is $|T_{P+1}|$. (Where P is an integer and f is a proper fraction, $0 < f < 1$)

ii) If $\frac{(n+1)|x|}{|x|+1} = P$ is an integer then there are **two** numerically greatest terms which are P^{th} and $(P+1)^{th}$ terms. Also $|T_P| = |T_{P+1}|$.

Note : To find numerically greatest term of $(a+b)^n$ we write $(a+b)^n = a^n(1+x)^n$ where $x = \frac{b}{a}$ and proceed.

13. i) If $n > 2$, $n \in \mathbb{N}$, then $(2n-1)^n + (2n)^n < (2n+1)^n$

ii) If the coefficients of x^{r-1} , x^r , x^{r+1} in $(1+x)^n$ are in A.P. then $(n-2r)^2 = n+2$.

14. i) $(1+\alpha)^n - 1$ is divisible by $M(\alpha)$

ii) $(1+\alpha)^n - n\alpha - 1$ is divisible by $M(\alpha^2)$

iii) $(1+\alpha)^n - {}^nC_2\alpha^2 - n\alpha - 1$ is divisible by $M(\alpha^3)$

15. i) Coefficient of x^{n-1} in $(x-\alpha_1)(x-\alpha_2)\dots(x-\alpha_n)$ is $-(\alpha_1 + \alpha_2 + \dots + \alpha_n)$

ii) Coefficient of x^{n-1} in $(x+\alpha_1)(x+\alpha_2)\dots(x+\alpha_n)$ is $(\alpha_1 + \alpha_2 + \dots + \alpha_n)$

iii) Coefficient of x^{n-2} in $(x-\alpha_1)(x-\alpha_2)\dots(x-\alpha_n)$ is $\frac{(\alpha_1 + \alpha_2 + \dots + \alpha_n)^2 - (\alpha_1^2 + \alpha_2^2 + \dots + \alpha_n^2)}{2}$

Binomial Coefficients :

$(x+a)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1}a + {}^nC_2 x^{n-2}a^2 + \dots + {}^nC_r x^{n-r}a^r + \dots + {}^nC_n a^n$, Here the coefficients ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_r, \dots, {}^nC_n$ are called binomial coefficients.

Note :

$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_r x^r + \dots + {}^nC_n x^n$ the coefficients ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_r, \dots, {}^nC_n$ are simply denoted by $C_0, C_1, C_2, \dots, C_r, \dots, C_n$ respectively

i.e., $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_r x^r + \dots + C_n x^n$

Standard results on Binomial coefficients

1. $C_0 + C_1 + C_2 + \dots + C_n = 2^n \Rightarrow \sum_{r=0}^n C_r = 2^n$

2. $C_0 - C_1 + C_2 - C_3 + \dots + (-1)^n C_n = 0$

3. $C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots = 2^{n-1}$

4. $a.C_0 + (a+d).C_1 + (a+2d).C_2 + \dots + (a+nd).C_n = (2a+nd) 2^{n-1}$

5. $a.C_0 - (a+d).C_1 + (a+2d).C_2 - \dots = 0$

6. $C_1 + 2.C_2 + 3.C_3 + \dots + n.C_n = n.2^{n-1} \Rightarrow \sum_{r=1}^n r.{}^nC_r = n.2^{n-1}$

7. $C_1 - 2.C_2 + 3.C_3 - \dots = 0 \Rightarrow \sum_{r=1}^n (-1)^{r-1} r.{}^nC_r = 0$

8. $a.C_0^2 + (a+d).C_1^2 + (a+2d).C_2^2 + \dots + (a+nd).C_n^2 = \frac{1}{2} (2a+nd) 2^n C_n$

9. ${}^m C_0 {}^n C_r + {}^m C_1 {}^n C_{r-1} + \dots + {}^m C_r {}^n C_0 = {}^{(m+n)} C_r$
10. $C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \dots + (C_0 + C_1 + \dots + C_{n-1}) = n \cdot 2^{n-1}$
11. $C_0 C_r + C_1 C_{r+1} + \dots + C_{n-r} C_n = 2^n C_{n-r}$ or $2^n C_{n+r}$
12. $C_0^2 + C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2 = 2^n C_n = \frac{(2n)!}{(n!)^2}$
13. $C_0^2 - C_1^2 + C_2^2 - \dots + (-1)^n C_n^2 = \begin{cases} 0 & \text{if } n \text{ is odd} \\ (-1)^{n/2} \cdot {}^n C_{n/2} & \text{if } n \text{ is even} \end{cases}$
14. $C_0 + \frac{C_1}{2}x + \frac{C_2}{3}x^2 + \dots + \frac{C_n}{n+1}x^n = \frac{(1+x)^{n+1} - 1}{(n+1)x}$
15. $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$
16. $\frac{C_1}{2} + \frac{C_3}{4} + \dots = \frac{2^n - 1}{n+1}$
17. $\frac{C_0}{1} - \frac{C_1}{2} + \frac{C_2}{3} - \dots + \frac{(-1)^n \cdot C_n}{n+1} = \frac{1}{n+1}$
18. $\frac{C_0}{1} + \frac{C_2}{3} + \frac{C_4}{5} + \dots = \frac{2^n}{n+1}$
19. a) $\sum_{r=0}^n r^2 \cdot C_r = n(n+1) \cdot 2^{n-2}$ b) $\sum_{r=0}^n (-1)^r r^2 \cdot C_r = 0$
20. a) $\sum_{r=0}^n r^3 \cdot C_r = n^2(n+3) \cdot 2^{n-3}$ b) $\sum_{r=0}^n (-1)^r r^3 \cdot C_r = 0$
21. Let $f(x)$ is any polynomial function which is expansion of any multinomial raised to some power,
 $f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$ is identity in x and they true for all real (or) complex
 $a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$
 - i) Sum of the Coefficients = $f(1)$
 - ii) Sum of the Coefficients of x having even powers is $\frac{f(1) + f(-1)}{2}$
 - iii) Sum of the Coefficients of x having odd powers is $\frac{f(1) - f(-1)}{2}$
 - iv) $a_0 - a_2 + a_4 + \dots = \frac{f(i) + f(-i)}{2}$

9. ${}^m C_0 {}^n C_r + {}^m C_1 {}^n C_{r-1} + \dots + {}^m C_r {}^n C_0 = {}^{(m+n)} C_r$
10. $C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \dots + (C_0 + C_1 + \dots + C_{n-1}) = n \cdot 2^{n-1}$
11. $C_0 C_r + C_1 C_{r+1} + \dots + C_{n-r} C_n = 2^n C_{n-r}$ or $2^n C_{n+r}$
12. $C_0^2 + C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2 = 2^n C_n = \frac{(2n)!}{(n!)^2}$
13. $C_0^2 - C_1^2 + C_2^2 - \dots + (-1)^n C_n^2 = \begin{cases} 0 & , \text{if } n \text{ is odd} \\ (-1)^{n/2} \cdot {}^n C_{\frac{n}{2}} & , \text{if } n \text{ is even} \end{cases}$
14. $C_0 + \frac{C_1}{2}x + \frac{C_2}{3}x^2 + \dots + \frac{C_n}{n+1}x^n = \frac{(1+x)^{n+1} - 1}{(n+1)x}$
15. $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$
16. $\frac{C_1}{2} + \frac{C_3}{4} + \dots = \frac{2^n - 1}{n+1}$
17. $\frac{C_0}{1} - \frac{C_1}{2} + \frac{C_2}{3} - \dots + \frac{(-1)^n \cdot C_n}{n+1} = \frac{1}{n+1}$
18. $\frac{C_0}{1} + \frac{C_2}{3} + \frac{C_4}{5} + \dots = \frac{2^n}{n+1}$
19. a) $\sum_{r=0}^n r^2 \cdot C_r = n(n+1) \cdot 2^{n-2}$ b) $\sum_{r=0}^n (-1)^r r^2 \cdot C_r = 0$
20. a) $\sum_{r=0}^n r^3 \cdot C_r = n^2(n+3) \cdot 2^{n-3}$ b) $\sum_{r=0}^n (-1)^r r^3 \cdot C_r = 0$
21. Let $f(x)$ is any polynomial function which is expansion of any multinomial raised to some power,
 $f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$ is identity in x and they true for all real (or) complex
 $a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$
- i) Sum of the Coefficients = $f(1)$
- ii) Sum of the Coefficients of x having even powers is $\frac{f(1) + f(-1)}{2}$
- iii) Sum of the Coefficients of x having odd powers is $\frac{f(1) - f(-1)}{2}$
- iv) $a_0 - a_2 + a_4 + \dots = \frac{f(i) + f(-i)}{2}$

$$\text{v) } a_0 + a_3 + a_6 + \dots = \frac{f(1) + f(w) + f(w^2)}{3}$$

$$\text{vi) } a_0 + a_4 + a_8 + \dots = \frac{f(1) + f(-1) + f(i) + f(-i)}{4}$$

$$\text{vii) } a_0 + a_n + a_{2n} + \dots = \frac{f(1) + f(\alpha) + f(\alpha^2) + \dots + f(\alpha^{n-1})}{n}$$

where $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ the n^{th} roots of unity

$$\text{viii) } (a_0 - a_2 + a_4 - \dots)^2 + (a_1 - a_3 + a_5 - \dots)^2 = f(i)f(-i)$$

Binomial theorem for rational Index :

If n is not a positive integer and $|x| < 1$ then

$$1. \quad (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r + \dots$$

$$2. \quad (1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \dots + \frac{n(n+1)(n+2)\dots(n+r-1)}{r!}x^r + \dots$$

$$3. \quad (1-x)^{-n} = 1 + {}^nC_1x + {}^{n+1}C_2x^2 + \dots + {}^{n+r-1}C_rx^r + \dots, \text{ if } n \text{ is a positive integer}$$

$$4. \quad (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + x^r + \dots + \infty$$

$$5. \quad (1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^r x^r + \dots + \infty$$

$$6. \quad (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \dots + (r+1)x^r + \dots + \infty$$

$$7. \quad (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots + (-1)^r (r+1)x^r + \dots + \infty$$

$$8. \quad (1-x)^{-3} = \frac{1}{1.2} [1.2 + 2.3x + 3.4x^2 + \dots + (r+1)(r+2)x^r + \dots + \infty]$$

$$9. \quad (1+x)^{-3} = \frac{1}{1.2} [1.2 - 2.3x + 3.4x^2 + \dots + (-1)^r (r+1)(r+2)x^r + \dots + \infty]$$

$$10. \quad (1-x)^{\frac{-p}{q}} = 1 + \frac{p}{q} \left(\frac{x}{q}\right) + \frac{p(p+q)}{2!} \left(\frac{x}{q}\right)^2 + \frac{p(p+q)(p+2q)}{3!} \left(\frac{x}{q}\right)^3 + \dots + \dots + \frac{p(p+q)(p+2q)\dots(p+(r-1)q)}{r!} \left(\frac{x}{q}\right)^r + \dots + \infty$$

$$11. \quad (1+x)^{\frac{-p}{q}} = 1 - \frac{p}{q} \left(\frac{x}{q}\right) + \frac{p(p+q)}{2!} \left(\frac{x}{q}\right)^2 - \frac{p(p+q)(p+2q)}{3!} \left(\frac{x}{q}\right)^3 + \dots + \dots + (-1)^r \frac{p(p+q)(p+2q)\dots(p+(r-1)q)}{r!} \left(\frac{x}{q}\right)^r + \dots + \infty$$

LECTURE SHEET

EXERCISE-I

Binomial expansion for positive integral index, Middle term, Numerically greatest term, R-f factor relation & Multinomial theorem

LEVEL-I (MAIN)

Single answer type questions

- The third term in the expansion of $\left(\frac{1}{x} + x \log_{10} x\right)^5$ is 1 then $x =$
 - 1
 - 10
 - 10^2
 - 10^4
- In the binomial expansion of $(a-b)^n$, $n > 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals
 - $\frac{n-4}{5}$
 - $\frac{5}{n-4}$
 - $\frac{6}{n-5}$
 - $\frac{n-5}{6}$
- If the ratio of the seventh term from beginning in $\left(\sqrt{2} + \frac{1}{\sqrt{3}}\right)^n$ to seventh term from end is $1/6$, then $n =$
 - 3
 - 6
 - 12
 - 9
- If the sum of odd terms and the sum of even terms in $(x+a)^n$ are p and q respectively then $4pq =$
 - $(x+a)^{2n} - (x-a)^{2n}$
 - $(x^2 - a^2)^{2n} + (x+a)^{2n}$
 - $(x^2 - a^2)^n - (x-a)^{2n}$
 - $(x^2 + a^2)^n + (x-a)^{2n}$
- If the sum of odd terms and the sum of even terms in $(x+a)^n$ are p and q respectively then $p^2 + q^2 =$
 - $\frac{(x+a)^{2n} - (x-a)^{2n}}{2}$
 - $(x+a)^{2n} - (x-a)^{2n}$
 - $\frac{(x+a)^{2n} + (x-a)^{2n}}{2}$
 - $(x+a)^{2n} + (x-a)^{2n}$
- If $T_0, T_1, T_2, \dots, T_n$ represent the terms in $(x+a)^n$, then

$$(T_0 - T_2 + T_4 - T_6 + \dots)^2 + (T_1 - T_3 + T_5 - \dots)^2$$
 is
 - $(x^2 - a^2)^2$
 - $(x^2 + a^2)^n$
 - $(a^2 - x^2)^n$
 - $(x^2 + a^2)^{2n}$
- The expression $\left[x + (x^3 - 1)^{1/2}\right]^5 + \left[x - (x^3 - 1)^{1/2}\right]^5$ is a polynomial of degree
 - 7
 - 4
 - 5
 - 6
- If the coefficient of $(2r+4)^{\text{th}}$ term and $(r-2)^{\text{th}}$ terms in the expansion of $(1+x)^{18}$ are equal then $r =$
 - 9
 - 4
 - 6
 - 3

9. If the coefficient of x^7 in $\left(ax^2 + \frac{1}{bx}\right)^{11}$ equals the coefficient of x^{-7} in $\left(ax - \frac{1}{bx^2}\right)^{11}$, then a and b satisfy the relation
- 1) $ab = 1$ 2) $\frac{a}{b} = 1$ 3) $a + b = 1$ 4) $a - b = 1$
10. In the expansion of $\left(\frac{1}{x^2} - x^3\right)^n$, $n \in N$, if the sum of the coefficients of x^5 and x^{10} is 0, then n is:
- 1) 25 2) 20 3) 15 4) None of these
11. If the constant term in the binomial expansion of $\left(x^2 - \frac{1}{x}\right)^n$, $n \in N$ is 15 then the value of n is equal to
- 1) 6 2) 9 3) 12 4) 15
12. Term independent of x in $\left(x - \frac{1}{x}\right)^4 \left(x + \frac{1}{x}\right)^3$ is
- 1) 1 2) 2 3) 0 4) 4
13. The term independent of x in the expansion of $(1+x)^n \left(1 + \frac{1}{x}\right)^n$ is
- 1) $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2$ 2) ${}^{2n}C_n$
 3) $\frac{1.3.5 \dots (2n-1)}{n!} 2^n$ 4) all the above
14. The coefficient of x^{51} in $\sum_{r=0}^{100} {}^{100}C_r (x-3)^{100-r} \cdot 2^r$ is
- 1) ${}^{100}C_{47}$ 2) ${}^{100}C_{53}$ 3) $-({}^{100}C_{53})$ 4) ${}^{100}C_{100}$
15. The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is
- 1) $(n-1)$ 2) $(-1)^n(1-n)$ 3) $(-1)^{n-1}(n-1)^2$ 4) $(-1)^{n-1}n$
16. Coefficient of x^5 in $(1+x)^{21} + (1+x)^{22} + \dots + (1+x)^{30}$ is
- 1) ${}^{51}C_5$ 2) 9C_5 3) ${}^{31}C_6 - {}^{21}C_6$ 4) ${}^{30}C_5 + {}^{20}C_5$
17. If r^{th} term is the middle term in the expansion of $\left(x^2 - \frac{1}{2x}\right)^{20}$, then $(r+3)^{\text{th}}$ term is:
- 1) ${}^{20}C_{14} \frac{x}{2^{14}}$ 2) ${}^{20}C_{12} x^2 2^{-12}$ 3) $-{}^{20}C_7 x^2 2^{-13}$ 4) none of these
18. The middle term in the expansion of $(1 - 3x + 3x^2 - x^3)^{2n}$ is
- 1) ${}^{6n}C_{3n} (-x)^{3n}$ 2) ${}^{6n}C_{2n} (-x)^{2n+1}$ 3) ${}^{4n}C_{3n} (-x)^{3n}$ 4) ${}^{6n}C_{3n} (-x)^{3n-1}$

19. The coefficient of the middle term in $(1 + \alpha x)^4$ and $(1 - \alpha x)^6$ is same then $\alpha =$
 1) $-5/3$ 2) $3/5$ 3) $-3/10$ 4) $10/3$
20. The numerically greatest term in the expansion $(5x - 6y)^{14}$ when $x = 2/5$, $y = 1/2$ is
 1) ${}^{14}C_6 \cdot 2^8 \cdot 3^6$ 2) ${}^{14}C_7 \cdot 2^6 \cdot 3^8$ 3) ${}^{14}C_6 \cdot 2^6 \cdot 3^8$ 4) ${}^{14}C_7 \cdot 2^8 \cdot 3^6$
21. The greatest coefficient (numerically) in $\left(2x - \frac{1}{3x}\right)^{10}$ is
 1) 5120 2) $\frac{1720}{3}$ 3) 1618 4) $\frac{5120}{3}$
22. If the middle term of $(1 + x)^{2n}$ is the greatest term then x lies between
 1) $n-1 < x < n$ 2) $\frac{n}{n+1} < x < \frac{n+1}{n}$ 3) $n < x < n+1$ 4) $\frac{n+1}{n} < x < \frac{n}{n+1}$
23. The greatest coefficient in $\left(\frac{x^{3/2}y}{2} + \frac{2}{xy^{3/2}}\right)^{12}$ is
 1) $12(2^{11})$ 2) $12(2^{10})$ 3) $12(2^{22})$ 4) $33(2^9)$
24. Integral part of $(7 + 4\sqrt{3})^n$ is ($n \in N$)
 1) an even number
 2) an odd number
 3) an even or an odd number depending upon the value of n
 4) nothing can be said
25. Integral part of $(7 + 5\sqrt{2})^{2n+1}$ is ($n \in N$)
 1) an even number
 2) an odd number
 3) an even or an odd number depending upon the value of n
 4) nothing can be said
26. If $R = (6\sqrt{6} + 14)^{2n+1}$ and $f = R - [R]$, where $[.]$ denotes the G.I.F., then $Rf =$
 1) 20^n 2) 20^{2n} 3) 20^{2n+1} 4) 1
27. The integral part of $(\sqrt{2} + 1)^6$ is
 1) 198 2) 196 3) 197 4) 199
28. The greatest integer which divides the number $101^{100} - 1$ is
 1) 10^2 2) 10^3 3) 10^4 4) 10^5
29. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is
 1) 2 2) 7 3) 8 4) 0
30. The coefficient of x^5 in the expansion of $(x^2 - x - 2)^5$ is
 1) -83 2) -82 3) -81 4) 0

31. Coefficient of $x^3y^4z^2$ in $(2x - 3y + 4z)^9$ is

- 1) $-\frac{9!}{4!4!}2^33^44^2$ 2) $-\frac{9!}{3!2!4!}2^33^44^2$ 3) $\frac{9!}{4!4!}2^33^44^2$ 4) $\frac{9!}{3!2!4!}2^33^44^2$

32. No. of terms in $(1 + 5\sqrt{2}x)^9 + (1 - 5\sqrt{2}x)^9$ if $x > 0$, is

- 1) 3 2) 5 3) 4 4) 6

33. No. of terms in $(1 + 3x + 3x^2 + x^3)^6$ is :

- 1) 17 2) 19 3) 21 4) 16

34. The number of distinct terms in $(a + b + c + d + e)^3$ is

- 1) 35 2) 38 3) 42 4) 45

35. No. of nonzero terms in $(1 + x)^{42} + (1 - x)^{42} + (1 + ix)^{42} + (1 - ix)^{42}$ is

- 1) 11 2) 12 3) 21 4) 22

36. The number of irrational terms in the expansion of $(\sqrt[5]{3} + \sqrt[3]{7})^{36}$ is

- 1) 30 2) 34 3) 31 4) 29

37. Sum of rational terms in $(\sqrt{2} + \sqrt[5]{3})^{10}$ is

- 1) 41 2) 42 3) 32 4) 38

38. $x + y = 1 \Rightarrow \sum_{r=0}^n r \cdot {}^nC_r x^r y^{n-r} =$

- 1) $2nx$ 2) nx 3) ny 4) nx^2

39. If the coefficients of $r, (r + 1), (r + 2)$ terms in $(1 + x)^{14}$ are in A.P. then $r =$

- 1) 3, 2 2) 5, 9 3) 2, 4 4) 5, 3

40. If a_1, a_2, a_3, a_4 are the coefficients of $2^{nd}, 3^{rd}, 4^{th}$ and 5^{th} terms of $(1 + x)^n$ respectively then

$\frac{a_1}{a_1 + a_2}, \frac{a_2}{a_2 + a_3}, \frac{a_3}{a_3 + a_4}$ are in

- 1) A.P. 2) G.P. 3) H.P. 4) A.G.P.

41. For natural numbers m, n if $(1 - y)^m(1 + y)^n = 1 + a_1y + a_2y^2 + \dots$, and $a_1 = a_2 = 10$, then (m, n) is :

- 1) (35, 20) 2) (45, 35) 3) (35, 45) 4) (20, 45)

Numerical value type questions

42. If 7 divides 32^{32} , then find the remainder

43. The sum $\sum_{i=0}^m \binom{10}{i} \binom{20}{m-i}$ (where $\frac{p}{q} = 0$, if $p < q$) is maximum when m is the sum of the digits of m is

44. If $f(x) = \sum_{r=1}^n [r^2({}^nC_r - {}^nC_{r-1}) + (2r+1){}^nC_r]$ and $f(30) = 30(2)^n$, then value of x is

LEVEL-II (ADVANCED)

Single answer type questions

- If the 4th term of $\left\{ \sqrt{x^{1+\log_{10} x}} + \sqrt[3]{x} \right\}^6$ is equal to 200, $x > 1$ and the logarithm is common logarithm, then x is not divisible is
 a) 2 b) 5 c) 10 d) 4
- If $p^4 + q^4 = 2$ ($p > 0, q > 0$), then the maximum value of term independent of x in the expansion of $\left(px^{\frac{1}{12}} + qx^{-\frac{1}{9}} \right)^{14}$ is
 a) ${}^{14}C_4$ b) ${}^{14}C_6$ c) ${}^{14}C_7$ d) ${}^{14}C_{12}$
- If $f(x)$ is periodic with period ' t ' such that $f(2x+3) + f(2x+7) = 2$, then the coefficient of m^{-3} in expansion of $\left(m + \frac{b}{m^3} \right)^{4t}$ is
 a) ${}^{16}C_7 b^7$ b) ${}^{32}C_{30} b^{30}$ c) ${}^{16}C_5 b^5$ d) ${}^{32}C_4 b^4$
- The coefficient of x^{501} in the expansion of $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \dots + x^{500}$ is
 a) ${}^{501}C_{301}$ b) ${}^{500}C_{301}$ c) ${}^{501}C_{300}$ d) none of these
- The coefficient of x^{70} in the product $(x-1)(x^2-2)(x^3-3)(x^4-4) \dots (x^{12}-12)$ is
 a) 4 b) 6 c) 8 d) 12
- Coefficient of x^{2016} in $(1+x+x^2+x^3+x^4)^{1001}(1-x)^{1002}(1+7x^{14})$ is
 a) 0 b) $-7 \cdot {}^{1001}C_{999}$ c) $7 \cdot {}^{1001}C_{403}$ d) ${}^{1001}C_{598}$
- Coefficient of x^6 in $((1+x)(1+x^2)^2(1+x^3)^3 \dots (1+x^n)^n)$ is
 a) 26 b) 28 c) 30 d) 35
- The sum of all the coefficients of those terms in the expansion of $(a+b+c+d)^8$ which contains b but not c is
 a) 6305 b) 6561 c) 256 d) 4^8
- The number of distinct terms in the expansion of $(x+y^2)^{13} + (x^2+y)^{14}$ is
 a) 27 b) 29 c) 28 d) 25
- If n is an even integer and a, b, c are distinct, the number of distinct terms in the expansion of $(a+b+c)^n + (a+b-c)^n$ is
 a) $\left(\frac{n}{2}\right)^2$ b) $\left(\frac{n+1}{2}\right)^2$ c) $\left(\frac{n+2}{2}\right)^2$ d) $\left(\frac{n+3}{2}\right)^2$

11. The coefficient of x^4 in the expansion of $\left(1 + 2x + \frac{3}{x^2}\right)^6$ is
 a) 240 b) 250 c) 260 d) 230
12. The number of terms in the expansion of $\left(x^3 + \frac{1}{x^3} + 1\right)^{100}$ is
 a) 301 b) 201 c) 101 d) None of these
13. The coefficient of $a^{10}b^7c^3$ in the expansion of $(bc + ca + ab)^{10}$ is
 a) 30 b) 60 c) 120 d) 240
14. If $x = (2 + \sqrt{3})^n$, $n \in N$ and $f = x - [x]$, then $\frac{f^2}{1-f}$ is
 a) an irrational number b) a non-integer rational number
 c) an odd number d) an even number
15. If $n > 0$ is an odd integer, and $x = (\sqrt{2} + 1)^n$ and $f = x - [x]$, then $\frac{1-f^2}{f}$ is
 a) an irrational number b) a non-integer rational number
 c) an odd integer d) an even integer
16. If 6th term in the expansion of $\left(\frac{3}{2} + \frac{x}{3}\right)^{11}$ is numerically greatest, when $x = 3$, then the sum of possible integral value of 'n' is
 a) 23 b) 24 c) 25 d) 26
17. The algebraically second largest term in the expansion of $(3 - 2x)^{15}$ at $x = \frac{4}{3}$.
 a) 5 b) 7 c) 9 d) 11
18. The remainder when $27^{10} + 7^{51}$ is divided by 10
 a) 4 b) 6 c) 9 d) 2

More than one correct answer type questions

19. The 9th term of $\left(\frac{\sqrt{10}}{(\sqrt{x})^{5 \log_{10} x}} + x \cdot x^{\frac{1}{2 \log_{10} x}}\right)^{10}$ is 450, then the rational value of x is
 a) 10 b) 100 c) $\frac{1}{10}$ d) $(10)^{-2/5}$
20. If a, b, c, d are any four consecutive coefficients of $(1+x)^n$ then which of the following is (are) correct
 a) $\frac{a}{a+b} + \frac{c}{c+d} = \frac{2b}{b+c}$ b) $\left(\frac{b}{b+c}\right)^2 > \frac{ac}{(a+b)(c+d)}$
 c) $\left(\frac{b}{b+c}\right)^2 < \frac{ac}{(a+b)(c+d)}$ d) $\left(\frac{b}{b+c}\right)^2 = \frac{ac}{(a+b)(c+d)}$

21. If recursion polynomials $P_k(x)$ are defined as $P_1(x) = (x-2)^2$, $P_2(x) = ((x-2)^2 - 2)^2$, $P_3(x) = (((x-2)^2 - 2)^2 - 2)^2$, ... (In general $P_k(x) = (P_{k-1}(x) - 2)^2$), then

- a) In $P_k(x)$ constant term is 4
 b) In $P_k(x)$ coefficient of x is 4^k
 c) In $P_k(x)$ coefficient of x is -4^k
 d) In $P_k(x)$ coefficient of x^2 is $\frac{4^{2k-1} - 4^{k-1}}{3}$

22. Which of the following statements is/are incorrect ?

- a) If $(3 + a\sqrt{2})^{100} + (3 + b\sqrt{2})^{100} = 7 + 5\sqrt{2}$. Number of pairs (a, b) for which the equation is true is one (a, b are rational numbers)
 b) The number of distance terms in the expansion of $\left(x^3 + \frac{1}{x^3} + 1\right)^{200}$ is 401
 c) In the expansion of $\left(\frac{x}{\cos\theta} + \frac{1}{x\sin\theta}\right)^{16}$. If I_1 is the least value of the term independent of x when $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$ and I_2 is the least value of the term independent of x when $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$ then $\frac{I_2}{I_1}$ is 16
 d) The sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$ is 1000.

Linked comprehension type questions

Passage - I :

Let $f(n) = 3^{2n} + 3^n + 1$ for every positive integer n . Answer the following questions:

23. Which of the following is true ?

- a) $f(n+3) = 3^6 f(n) - 702 \cdot 3^n - 728$
 b) $f(n+3) = 3^6 f(n) - 701 \cdot 3^n - 729$
 c) $f(n+3) = 3^6 f(n) + 702 \cdot 3^n - 728$
 d) None of these

24. Which of the following is false ?

- a) $f(100)$ is divisible by 13
 b) $f(1001)$ is divisible by 13
 c) $f(2007)$ is divisible by 13
 d) None of these

25. Which of the following is true?

- a) $f(50)$ leaves remainder 1 when divided by 13
 b) $f(51)$ leaves remainder 0 when divided by 13
 c) $f(51)$ leaves remainder 3 when divided by 13
 d) None of these

Passage - II :

To find coefficient of x^r ($0 \leq r \leq n-1$) in the expansion

$(x+a)^{n-1} + (x+a)^{n-2}(x+b) + \dots + (x+a)(x+b)^{n-2} + (x+b)^{n-1}$ we first sum up the series

26. The coefficient of x^r ($0 \leq r \leq n-1$) in the expansion

$E = (x+2)^{n-1} + (x+2)^{n-2}(x+1) + (x+2)^{n-3}(x+1)^2 + \dots + (x+1)^{n-1}$

- a) nC_r
 b) ${}^nC_r(2^{n-r} - 1)$
 c) ${}^nC_r 2^{n-r}$
 d) none of these

EXERCISE-II

Properties of Binomial coefficients, Summation of series using multinomial coefficients & Multiple summations

LEVEL-I (MAIN)

Single answer type questions

- $C_0 + 4 \cdot C_1 + 7 \cdot C_2 + \dots + (n+1) \text{ terms} =$
 - $(3n+2) \cdot 2^{n-1}$
 - $(2n+2) \cdot 2^{n-1}$
 - $(2n+2) \cdot 3^{n-1}$
 - $(2n-2) \cdot 3^{n+1}$
- $3 \cdot C_0 - 7 \cdot C_1 + 11 \cdot C_2 - \dots + (n+1) \text{ terms} =$
 - 0
 - 1
 - 1
 - 2
- $$\frac{(C_0 + C_1)(C_1 + C_2)(C_2 + C_3) \dots (C_{n-1} + C_n)}{C_0 C_1 C_2 \dots C_n} =$$
 - $\frac{(n+1)^n}{n!}$
 - $\frac{n+1}{n!}$
 - $\frac{(n+1)^{n-1}}{n!}$
 - $\frac{(n-1)^n}{n!}$
- $$\frac{{}^n C_1 + {}^{(n+1)} C_2 + {}^{(n+2)} C_3 + \dots + {}^{(n+m-1)} C_m}{{}^m C_1 + {}^{(m+1)} C_2 + {}^{(m+2)} C_3 + \dots + {}^{(m+n-1)} C_n} =$$
 - 1
 - 2
 - 3
 - 4
- $$\sum_{r=0}^{n-1} \frac{C_r}{C_r + C_{r+1}} =$$
 - $\frac{n}{2}$
 - $\frac{n}{3}$
 - $\frac{n}{4}$
 - $\frac{2n}{3}$
- Sum of last 8 coefficients in $(1+x)^{16}$ is ____
 - $\left[2^{15} - \frac{1}{2} \cdot {}^{16} C_8 \right]$
 - $\left[2^{15} + \frac{1}{2} \cdot {}^{16} C_2 \right]$
 - $\left[2^{15} - \frac{1}{2} \cdot {}^{16} C_2 \right]$
 - $\left[2^{15} - \frac{1}{4} \cdot {}^{16} C_2 \right]$
- ${}^{(2n+1)} C_0 - {}^{(2n+1)} C_1 + {}^{(2n+1)} C_2 - \dots + {}^{(2n+1)} C_{2n} =$
 - 0
 - 1
 - 1
 - 2
- ${}^{15} C_2 + 2 \cdot {}^{15} C_3 + 3 \cdot {}^{15} C_4 + \dots + 14 \cdot {}^{15} C_{15} =$
 - $13 \cdot 2^{14} - 1$
 - $13 \cdot 2^{14} + 1$
 - $12^{14} + 1$
 - $12^{14} - 1$
- $C_0 - [C_1 - 2 \cdot C_2 + 3 \cdot C_3 - \dots + (-1)^{n-1} \cdot n \cdot C_n] =$
 - 0
 - 1
 - 1
 - 2
- $2 \cdot C_2 + 6 \cdot C_3 + 12 \cdot C_4 + \dots + n(n-1) \cdot C_n =$
 - $n(n-1) \cdot 2^{n-1}$
 - $2n(n-1) \cdot 2^{n-2}$
 - $n(n-1) \cdot 2^{n-2}$
 - $2n(n+1) \cdot 2^{n-1}$

11. $\frac{C_0}{1} + \frac{C_2}{3} + \frac{C_4}{5} + \dots + \frac{C_{16}}{17} =$

1) $\frac{2^{15}}{14}$

2) $\frac{2^{16}}{17}$

3) $\frac{2^{15}}{16}$

4) $\frac{2^{20}}{22}$

12. $\frac{C_1}{2} + \frac{C_3}{4} + \dots + \frac{C_{15}}{16} =$

1) $\frac{2^{15}-1}{16}$

2) $\frac{2^{15}+1}{16}$

3) $\frac{2^{14}+1}{16}$

4) $\frac{2^{20}+1}{16}$

13. $\frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \dots + \frac{C_n}{n+2} =$

1) $\frac{2n \cdot 2^{n+1} - 1}{(n-1)(n-2)}$

2) $\frac{n \cdot 2^{n+1} - 1}{(n-1)(n-2)}$

3) $\frac{n \cdot 2^{n+1} - 1}{(n+1)(n-2)}$

4) $\frac{n \cdot 2^{n+1} + 1}{(n+1)(n+2)}$

14. $\frac{C_0}{2} + \frac{C_1}{6} + \frac{C_2}{12} + \dots + \frac{C_n}{(n+1)(n+2)} =$

1) $\frac{2^{n+2} - n - 2}{(n+1)(n+2)}$

2) $\frac{2^{n+2} - n - 3}{(n+1)(n+2)}$

3) $\frac{2^{n+1} - n - 3}{(n+1)(n+2)}$

4) $\frac{2^{n+1} - n - 3}{(n-1)(n-2)}$

15. $C_1^2 + 2 \cdot C_2^2 + 3 \cdot C_3^2 + \dots + n \cdot C_n^2 =$

1) $n \cdot 2^n C_n$

2) $\frac{n}{2} \cdot 2^n C_{n-1}$

3) $\frac{n}{2} \cdot 2^n C_n$

4) $\frac{n}{2} \cdot 2^n C_{n+1}$

16. If ${}^{2n}C_r = C_r$, then $C_1^2 - 2 \cdot C_2^2 + 3 \cdot C_3^2 - 4 \cdot C_4^2 + \dots + 2n \cdot C_{2n}^2 =$

1) $\frac{(-1)^{n-1} \cdot (2n)!}{(n-1)!}$

2) $\frac{(-1)^n \cdot (2n)!}{(n+1)!}$

3) $\frac{(-1)^{n-1} \cdot (2n)!}{n!(n-1)!}$

4) $\frac{(-1)^n \cdot (2n)!}{(n+1)!n!}$

17. $C_0 C_2 + C_1 C_3 + C_2 C_4 + \dots + C_{n-2} C_n =$

1) $2^n C_{n-2}$

2) $2^n C_n$

3) $2^n C_{n-1}$

4) $2^n C_{2n-2}$

18. $({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + ({}^{2n}C_{2n})^2 =$

1) $(-1)^n \cdot 2^n C_n$

2) $(-1)^{2n} \cdot 2^n C_n$

3) $(-1)^n \cdot 3^n C_n$

4) $(-1)^n \cdot {}^n C_n$

19. ${}^{2n+1}C_0^2 - {}^{2n+1}C_1^2 + {}^{2n+1}C_2^2 - \dots - {}^{2n+1}C_{2n+1}^2 =$

1) 0

2) $({}^{2n+1}C_n)$

3) $-({}^{2n+1}C_n)$

4) $-\frac{1}{2}({}^{2n}C_n)$

20. $C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \dots + (C_0 + C_1 + C_2 + \dots + C_n) = (n \text{ is even})$

1) $(n+2)2^{n-1}$

2) $(n+1)2^{n-1}$

3) $(n-2)2^{n-1}$

4) $(n-2)2^{n+1}$

21. If ${}^{10}C_1 \cdot {}^9C_5 + {}^{10}C_2 \cdot {}^9C_4 + {}^{10}C_3 \cdot {}^9C_3 + {}^{10}C_4 \cdot {}^9C_2 + {}^{10}C_5 \cdot {}^9C_1 + {}^{10}C_6 = {}^{19}C_6 + x$ then $x =$

- 1) -84 2) 84 3) 81 4) -81

22. $\sum_{r=1}^n (-1)^{r-1} {}^nC_r (a-r) =$

- 1) a 2) $-a$ 3) $2a$ 4) $3a$

23. $\sum_{r=0}^n \frac{1}{{}^nC_r} = S_n$, $t_n = \sum_{r=0}^n \frac{r}{{}^nC_r}$ then $\frac{t_n}{S_n} =$

- 1) $\frac{1}{4}n$ 2) $\frac{1}{3}n$ 3) $\frac{1}{2}n$ 4) n

24. Sum of coefficients of terms of even powers of x in $(1+x+x^2+x^3)^8$ is

- 1) 512 2) 516 3) 612 4) 234

25. Sum of coefficients of terms of odd powers of x in $(1+x-x^2-x^3)^8$ is

- 1) 0 2) 1 3) 2 4) -1

26. Sum of coefficients of all the integral powers of x in $(1+2\sqrt{x})^{40}$ is

- 1) $\frac{3^{40}-1}{2}$ 2) $\frac{3^{40}+1}{2}$ 3) $\frac{3^{38}-1}{2}$ 4) $\frac{3^{38}+1}{2}$

27. If the sum of all the Binomial coefficients in $(x+y)^n$ is 512, then the greatest Binomial coefficient is

- 1) ${}^{10}C_5$ 2) 9C_4 or 9C_5 3) ${}^{11}C_5$ or ${}^{11}C_6$ 4) ${}^{12}C_6$

28. $(1+2x+3x^2)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{20}x^{20} \Rightarrow \frac{a_2}{a_1} =$

- 1) 10.5 2) 21 3) 10 4) 5.5

29. If a_r is the coefficient of x^r in the expansion of $(1+x+x^2)^n$ then $a_1 - 2a_2 + 3a_3 - \dots - 2na_{2n} =$

- 1) 0 2) n 3) $-n$ 4) $2n$

30. If a_k is the coefficient of x^k in the expansion of $(1+x+x^2)^n$ for $k = 0, 1, 2, \dots, 2n$ then

$$a_1 + 2a_2 + 3a_3 + \dots + 2n \cdot a_{2n} =$$

- 1) $-a_0$ 2) 3^n 3) $n \cdot 3^n$ 4) $-n \cdot 3^n$

31. $(1+x+x^2+\dots+x^p)^n = a_0 + a_1x + a_2x^2 + \dots + a_{np}x^{np} \Rightarrow a_1 + 2a_2 + 3a_3 + \dots + np \cdot a_{np} =$

- 1) $\frac{np(p+1)^n}{2}$ 2) $\frac{np(p+1)^n}{4}$ 3) $\frac{np(p-1)^n}{4}$ 4) $\frac{np(p-1)^{2n}}{4}$

32. If $(1+x-2x^2)^8 = 1 + a_1x + a_2x^2 + \dots + a_{16}x^{16}$, then $a_2 + a_4 + a_6 + \dots + a_{16} =$

- 1) 120 2) 123 3) 127 4) 231

33. If $(1+x-2x^2)^8 = 1 + a_1x + a_2x^2 + \dots + a_{16}x^{16}$, then $a_1 + a_3 + a_5 + \dots + a_{15} =$
 1) 2^7 2) -2^7 3) 3^2 4) 4^6
34. If $(1+x+x^2)^8 = a_0 + a_1x + \dots + a_{16}x^{16}$ then $a_0 - a_2 + a_4 - a_6 + \dots + a_{16} =$
 1) 1 2) 2 3) 3 4) 4
35. If $(1+x+x^2)^8 = a_0 + a_1x + \dots + a_{16}x^{16}$ then $a_1 - a_3 + a_5 - a_7 + \dots - a_{15} =$
 1) 1 2) 2 3) 3 4) 0
36. If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$ then $a_0 + a_3 + a_6 + \dots =$
 1) 3^{n-1} 2) 0 3) 1 4) 3^n
37. If $(1+x)^{10} = \sum_{r=0}^{10} C_r x^r$ then $(C_0 - C_2 + C_4 - C_6 + C_8 - C_{10})^2 + (C_1 - C_3 + C_5 - C_7 + C_9)^2 =$
 1) 2^{10} 2) -2^{10} 3) 2^{12} 4) -2^{12}
38. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then $C_0 + C_4 + C_8 + \dots =$
 1) $2^{n-2} + 2^{\frac{n-1}{2}} \cos \frac{n\pi}{4}$ 2) $2^{n-2} + 2^{\frac{n-1}{2}} \sin \frac{n\pi}{4}$ 3) $2^{n-1} + 2^n \cos \frac{n\pi}{4}$ 4) $2^{n-1} + 2^{\frac{n}{2}} \sin \frac{n\pi}{4}$
39. $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{a^{i+j+k}}$ is equal to (where $|a| > 1$)
 1) $(a-1)^{-3}$ 2) $\frac{3}{a-1}$ 3) $\frac{1}{a^3-1}$ 4) None of these
40. The value of $\sum_{1 \leq i \leq j \leq k \leq w \leq n} 1$ is
 1) ${}^nC_4 + {}^nC_3$ 2) ${}^nC_4 + {}^nC_3 + {}^nC_2$ 3) ${}^nC_4 + 2{}^nC_3 + {}^nC_2$ 4) ${}^nC_4 + {}^nC_3 + 2{}^nC_2$

LEVEL-II (ADVANCED)

Single answer type questions

1. Consider the sequence $\frac{{}^nC_0}{1.2.3}, \frac{{}^nC_1}{2.3.4}, \frac{{}^nC_2}{3.4.5}, \dots$, if $n = 50$ then greatest term is
 a) 30^{th} b) 24^{th} c) 26^{th} d) 27^{th}
2. If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$ where ${}^nC_0, {}^nC_1, {}^nC_2, \dots$ are binomial coefficients, then $2(C_0 + C_3 + C_6 + \dots) + (C_1 + C_4 + C_7 + \dots)(1+\omega) + (C_2 + C_5 + C_8 + \dots)(1+\omega^2)$, where ω is the cube root of unity and n is a multiple of 3, then the above expression is equal to
 a) $2^n + 1$ b) $2^{n-1} + 1$ c) $2^{n+1} - 1$ d) $2^n - 1$

3. Value of $\frac{\sum_{k=0}^r {}^nC_{2k} {}^{n-2k}C_{r-k}}{\sum_{k=0}^n {}^nC_k {}^{2k}C_{2r} \left(\frac{3}{4}\right)^{n-k} \left(\frac{1}{2}\right)^{2k-2r}} (n \geq 2r)$ is
- a) $1/2$ b) 2 c) 1 d) None of these
4. The sum of the series $\sum_{r=1}^{3n-1} \frac{(-1)^{r-1} r}{{}^{3n}C_r}$ is (where n is an even natural number)
- a) 0 b) $\frac{3n}{3n+1}$ c) $\frac{3n+1}{3n+2}$ d) $\frac{3n}{3n+2}$
5. If $C_0, C_1, C_2, \dots$ are binomial coefficients in the expansion $\sum_{r=0}^n C_r x^r$, then value of the expression (series) $\frac{2C_0}{1} + \frac{3C_1}{2} + \frac{4C_2}{3} + \frac{5C_3}{4} + \dots$ is
- a) $\frac{2^n+1}{n+1}$ b) $\frac{2^n-1}{n+1}$ c) $\frac{2^n(n+3)-1}{n+1}$ d) $\frac{2^n(n+2)-1}{n+1}$
6. Given ${}^8C_1 x(1-x)^7 + 2^8 \cdot {}^8C_2 x^2(1-x)^6 + 3^8 \cdot {}^8C_3 x^3(1-x)^5 + \dots + 8^8 x^8 = a_0 + a_1 x + a_2 x^2 + \dots + a_8 x^8$ then $a_0 + a_1$ is
- a) 6 b) 5 c) 8 d) 9
7. If $\sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r$ and $a_k = 1 \forall k \geq n$ then $b_n =$
- a) ${}^{2n}C_n$ b) ${}^{2n+1}C_{n+1}$ c) nC_0 d) ${}^{2n}C_{n+1}$
8. If $n \in N$, then $\sum_{r=0}^n (-1)^r \cdot {}^nC_r \cdot \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \dots \text{to } m \text{ terms} \right]$
- a) $\frac{2^{mn}}{(2^n-1)2^m}$ b) $\frac{2^{mn}+1}{(2^n-1)2^{mn}}$ c) $\frac{2^{mn}-1}{(2^n-1)2^{mn}}$ d) None
9. In a ΔABC $\sum_{r=0}^n {}^nC_r \cdot a^{n-r} \cdot b^r \cos(rA - (n-r)B) =$
- a) $(a+b)^n$ b) $(a-b)^n$ c) c^n d) c^{2n}
10. The largest integer k such that 3^k divides $2^{3^n} + 1, n \in N$ is
- a) 2 b) n c) $n-1$ d) $n+1$
11. If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$ then $6(a_0 + a_6 + a_{12} + a_{18} + \dots) =$
- a) $3^n - 1 + 2^{n+1} \cos \frac{n\pi}{3}$ b) $3^n + 1 + 2^{n+1} \cdot \cos \frac{n\pi}{3}$
c) $3^n - 1 + 2^n \sin \frac{n\pi}{3}$ d) $3^n + 1 + 2^n \sin \frac{n\pi}{3}$

12. The value of $C_3 + C_7 + C_{11} + \dots$ is

a) $\frac{1}{2} \left(2^{n-1} - 2^{n/2} \sin \frac{n\pi}{4} \right)$

b) $\frac{1}{2} \left(2^{n-1} + 2^{n/2} \sin \frac{n\pi}{4} \right)$

c) $\frac{1}{4} \left(2^{n+1} - 2^{n/2} \sin \frac{n\pi}{4} \right)$

d) none

13. If k and n are positive integers and $S_k = 1^k + 2^k + 3^k + \dots + n^k$ then $\sum_{r=1}^m {}^{m+1}C_r (S_r)$ is equal to

a) $(n+1)^{m+1} - (n+1)$

b) $(n+1)^{m+1} + (n+1)$

c) $(n-1)^{m+1} - (n-1)$

d) none

14. The sum of all the coefficients of those terms in the expansion of $(a + b + c + d)^8$ which contains b but not c is

a) 6305

b) 6561

c) 256

d) 4^8

15. If $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$ then $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2$ is equals

a) $\frac{1}{2}a_n$

b) $\frac{1}{2}a_{n+1}$

c) a_{n+1}

d) a_n

16. The value of $\sum_{0 \leq i < j \leq n} i({}^nC_j)$ is equal to

a) $n(n+1)2^{n-3}$

b) $n^2 2^{n-3}$

c) $n(n-1)2^{n-3}$

d) none

17. The value of the expression $\sum_{0 \leq i < j \leq n} (-1)^{i+j-1} {}^nC_i \cdot {}^nC_j =$

a) $2^{n-1} C_n$

b) $2^n C_n$

c) $2^{n+1} C_n$

d) none

18. Let $S_1 = \sum_{0 \leq i < j \leq 100} C_i C_j$, $S_2 = \sum_{0 \leq j < i \leq 100} C_i C_j$ and $S_3 = \sum_{0 \leq i = j \leq 100} C_i C_j$ where C_i represents coefficient of x^i in the binomial expansion of $(1+x)^{100}$. If $S_1 + S_2 + S_3 = a^b$ where $a, b \in \mathbb{N}$, then the least value of $(a+b)$ is

a) 66

b) 72

c) 46

d) 52

More than one correct answer Type Questions

19. ${}^nC_0 {}^{2n}C_m - {}^nC_1 {}^{2n-2}C_m + {}^nC_2 {}^{2n-4}C_m - \dots =$

a) $\binom{n}{m-n} 2^{2n-m}$ if $m \geq n$

b) 0 if $m < n$

c) $\binom{n}{m-n} 2^{2n+m}$ if

d) 1 if

20. Which of the following is/are correct?

a) ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - \dots - {}^{20}C_{15} = -{}^{19}C_{15}$

b) ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - \dots - {}^{20}C_{15} = -{}^{20}C_{14}$

c) $16 {}^{20}C_0 - 15 {}^{20}C_1 + 14 {}^{20}C_2 - \dots - 2 {}^{20}C_{14} - {}^{20}C_{15} = {}^{19}C_{14}$

d) $16 {}^{20}C_0 - 15 {}^{20}C_1 + 14 {}^{20}C_2 - \dots - 2 {}^{20}C_{14} - {}^{20}C_{15} = {}^{18}C_{15}$

21. Let $(1 + \sqrt{2})^n = x_n + y_n\sqrt{2}$ where x_n, y_n are integers, then

- a) $x_n^2 - 2y_n^2 = (-1)^n$ b) $x_n + 2y_n - x_{n+1} = 0$ c) $x_n^2 - 2y_n^2 = 1$ d) $y_{n+1} = x_n + y_n$

22. If $(x^{2006} + x^{2008} + 2)^{2010} = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ then value of

$$a_0 - \frac{1}{2}a_1 - \frac{1}{2}a_2 + a_3 - \frac{1}{2}a_4 - \frac{1}{2}a_5 + a_6 + \dots \text{ is}$$

- a) less than 2 b) greater than 0 c) equals 2 d) none of these

23. If $ac > b^2$, then the sum of the coefficients in the expansion of $(a\alpha^2x^2 + 2b\alpha x + c)^n$ is, where $a, b, c, \alpha \in \mathbb{R}$ and $n \in \mathbb{N}$

- a) positive if $a > 0$ b) positive if $c > 0$
c) negative if $a < 0$ and n is odd d) positive if $c < 0$ and n is even

Linked comprehension type questions

Passage - I :

$(1 + ax + bx^2 + cx^3)^{10} = 1 + p_1x + p_2x^2 + p_3x^3 + \dots + p_{30}x^{30}$ And the values of p_1, p_2, p_3 respectively are 20, 200, 1000 respectively then

24. $b =$

- a) -1 b) 2 c) 3 d) 4

25. $c =$

- a) 12 b) -15 c) +30 d) -32

26. $p_4 =$

- a) 540 b) 600 c) 660 d) 480

Passage - II :

Let $(1+x)^{20} = \sum_{r=0}^{20} a_r x^r$ when $a_r = {}^{20}C_r$ Then

27. $\sum_{0 \leq i < j \leq 20} a_i a_j =$

- a) $2^{40} - {}^{40}C_{20}$ b) $2^{39} - {}^{40}C_{20}$ c) $2^{39} - {}^{39}C_{20}$ d) $2^{40} - {}^{39}C_{19}$

28. $\sum_{0 \leq i < j \leq 20} (a_i - a_j)^2 =$

- a) $42 \cdot {}^{39}C_{19} - 2^{40}$ b) $21 \cdot {}^{40}C_{20} - 2^{39}$ c) $21 \cdot {}^{39}C_{19} - 2^{40}$ d) $21 \cdot {}^{40}C_{20} - 2^{40}$

29. $\sum_{0 \leq i < j \leq 20} (i+j) a_i a_j$

- a) $40(2^{39} - {}^{39}C_{20})$ b) $20(2^{39} - {}^{39}C_{20})$ c) $40(2^{40} - {}^{40}C_{20})$ d) $40(2^{40} - {}^{39}C_{19})$

Matrix matching type question

30. COLUMN - I

COLUMN - II

A) $\left({}^{32}C_0\right)^2 - \left({}^{32}C_1\right)^2 + \left({}^{32}C_2\right)^2 = \text{---}$

p) ${}^{61}C_{32}$

B) $\left({}^{32}C_0\right)^2 + \left({}^{32}C_1\right)^2 + \left({}^{32}C_2\right)^2 + \dots + \left({}^{32}C_{32}\right)^2 = \text{---}$

q) 0

C) $\frac{1}{32} \left(1 \left({}^{32}C_1\right)^2 + 2 \left({}^{32}C_2\right)^2 + 3 \left({}^{32}C_3\right)^2 + \dots + 32 \left({}^{32}C_{32}\right)^2 \right) = \text{---}$

r) ${}^{32}C_{16}$

D) $\left({}^{31}C_0\right)^2 - \left({}^{32}C_1\right)^2 + \left({}^{32}C_2\right)^2 - \left({}^{31}C_3\right)^2 + \dots + \left({}^{31}C_{31}\right)^2 = \text{---}$

s) ${}^{64}C_{32}$

31. Let $A = \sum_{r=1}^{50} \frac{{}^{50+r}C_r (2r-1)}{{}^{50}C_r (50+r)}$, $B = \sum_{r=1}^{50} \left({}^{50}C_r\right)^2$, $C = \sum_{r=1}^{100} (-1)^r \left({}^{100}C_r\right)^2$ then match the following

COLUMN - I

COLUMN - II

A) $A - B = \text{---}$

p) -1

B) $C - A = \text{---}$

q) 0

C) $B - A = \text{---}$

r) 1

D) $B + C - 2A = \text{---}$

s) 2

Integer answer type questions

32. If ${}^{2015}C_1 - {}^{2015}C_2 \left(1 + \frac{1}{2}\right) + {}^{2015}C_3 \left(1 + \frac{1}{2} + \frac{1}{3}\right) - \dots + {}^{2015}C_{2015} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2015}\right) = K$, then the sum of the digits of $\left[\frac{1}{k}\right]$, where $[.]$ denotes G.I.F is _____

33. Let $a = 3^{\frac{1}{223}} + 1$ and for all $n \geq 3$, $f(n) = {}^nC_0 a^{n-1} - {}^nC_1 a^{n-2} + {}^nC_2 a^{n-3} - \dots + (-1)^{n-1} {}^nC_{n-1} a^0$. If the value of $f(2007) + f(2008) = 3^k$ where $k \in \mathbb{N}$, then $k =$

34. ${}^nC_0 \cdot {}^{2n}C_n - {}^nC_1 \cdot {}^{(2n-1)}C_n + {}^nC_2 \cdot {}^{(2n-2)}C_n + \dots + (-1)^n \cdot {}^nC_n \cdot {}^nC_n =$

35. If S be the sum of coefficients in the expansion of $(px + qy - rz)^n$ (where $p, q, r > 0$), then the value

of $\lim_{n \rightarrow \infty} \frac{S}{(S^{1/n} + 1)^n}$ is

36. $C_0, C_1, C_2, \dots, C_n$ are binomial coefficients in the expansion of $(1+x)^n$ then

$\lim_{n \rightarrow \infty} \left\{ C_n - C_{n-1} \left(\frac{2}{3}\right) + C_{n-2} \left(\frac{2}{3}\right)^2 - \dots + (-1)^n C_0 \left(\frac{2}{3}\right)^n \right\} =$

37. In the expansion of $(1+3x+2x^2)^6$, then coefficient of x^{11} is $k \times 2^6$, then k is

38. If for $1 \leq m \leq n$, $f(m, n) = {}^nC_0 - {}^nC_1 + {}^nC_2 - \dots + (-1)^{m-1} \cdot {}^nC_{m-1}$, then $f(7, 8)$ is

EXERCISE-III

*Binomial theorem for rational index, Approximations &
Summation of series using multinomial*

LEVEL-I (MAIN)

Single answer type questions

Rational index :

- The range of x so that the expansion of $(3-4x)^{1/2}$ is valid is
 1) $-3/4 < x < 3/4$ 2) $|x| < 3$ 3) $|x| < 1/4$ 4) $|x| < 1$
- If the expansion $(4a-8x)^{1/2}$ were to be possible then
 1) $2 < \left| \frac{a}{x} \right|$ 2) $2 > \left| \frac{a}{x} \right|$ 3) $2 < \left| \frac{x}{a} \right|$ 4) $2 > \left| \frac{x}{a} \right|$
- For $|x| > \frac{3}{2}$, the value of the third term in the expansion of $(3+2x)^{3/5}$ is
 1) $\frac{27}{50} \cdot 2^{\frac{3}{5}} \cdot x^{\frac{9}{5}}$ 2) $\frac{27}{50} \cdot 2^{\frac{3}{5}} \cdot x^{\frac{7}{5}}$ 3) $\frac{27}{50} \cdot 2^{\frac{2}{5}} \cdot x^{\frac{7}{5}}$ 4) $-\frac{27}{50} \cdot 2^{\frac{2}{5}} \cdot x^{\frac{7}{5}}$
- If $\frac{1}{(1-2x)(1+3x)}$ is to be expanded as a power series of x , then
 1) $|x| < 1/2$ 2) $|x| < 1/6$ 3) $-1/3 < x < 1/2$ 4) $|x| < 1/3$
- $1 + {}^2C_1 x + {}^3C_2 x^2 + {}^4C_3 x^3 + \dots$ to ∞ terms can be summed up if
 1) $x < 1$ 2) $x > -1$ 3) $-1 < x < 1$ 4) $-\infty < x < \infty$
- For $|x| < 1$, the $(r+1)^{\text{th}}$ term in the expansion of $\sqrt{1-x}$ is
 1) $\frac{1 \cdot 3 \cdot 5 \dots (2r-3)}{r!} \left(\frac{x}{2}\right)^r$ 2) $-\frac{1 \cdot 3 \cdot 5 \dots (2r-3)}{r!} \left(\frac{x}{2}\right)^r$
 3) $-\frac{1 \cdot 3 \cdot 5 \dots (2r-3)}{r!} (x)^r$ 4) $\frac{1 \cdot 3 \cdot 5 \dots (2r-3)}{r!} (x)^r$
- The general term of $(2a-3b)^{-1/2}$ is
 1) $\frac{1 \cdot 3 \cdot 5 \dots (2r-5)}{r!} \frac{1}{\sqrt{2a}} \left(\frac{3b}{4a}\right)^r$ 2) $\frac{1 \cdot 3 \cdot 5 \dots (2r-3)}{r!} \frac{1}{\sqrt{2a}} \left(\frac{3b}{4a}\right)^r$
 3) $\frac{1 \cdot 3 \cdot 5 \dots (2r-1)}{r!} \frac{1}{\sqrt{2a}} \left(\frac{3b}{4a}\right)^r$ 4) $\frac{1 \cdot 3 \cdot 5 \dots (2r-3)}{r!} \frac{1}{\sqrt{a}} \left(\frac{3b}{4a}\right)^r$
- If $|x| < 1$, then the coefficient of x^n in expansion of $(1+x+x^2+x^3+\dots)^2$ is
 1) n 2) $n-1$ 3) $n+2$ 4) $n+1$

9. The coefficient of x^{24} in $(1 + 3x + 6x^2 + 10x^3 + \dots \infty)^{23}$ is
 1) 25 2) 125 3) 50 4) 300
10. If S_n denotes the sum of first n natural numbers then $S_1 + S_2 x + S_3 x^2 + \dots + S_n x^{n-1} + \dots \infty$ terms =
 1) $(1-x)^{-1}$ 2) $(1-x)^{-2}$ 3) $(1-x)^{-3}$ 4) $(1-x)^{-4}$
11. ${}^4C_1 + {}^5C_2 \cdot \left(\frac{1}{2}\right) + {}^6C_3 \cdot \left(\frac{1}{2}\right)^2 + \dots$ to ∞ terms
 1) 30 2) 40 3) 900 4) 15
12. The coefficient of x^2 in $(1+x)^2(8-x)^{-1/3}$ is
 1) $\frac{2167}{4032}$ 2) $\frac{2265}{4132}$ 3) $\frac{313}{576}$ 4) $\frac{3691}{6792}$
13. The coefficient of x^n in $(1+x)^n \left(1+\frac{1}{x}\right)^n$ is
 1) 0 2) 1 3) 2^n 4) $2^n C_n$
14. If $|x| < \frac{1}{2}$, then the coefficient of x^r in the expansion of $\frac{1+2x}{(1-2x)^2}$ is
 1) $r \cdot 2^r$ 2) $(2r-1) 2^r$ 3) $r \cdot 2^{2r+1}$ 4) $(2r+1) 2^r$
15. If the expansion in powers of x of the function $\frac{1}{(1-ax)(1-bx)}$ is $a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$ then a_n is :
 1) $\frac{a^n - b^n}{b-a}$ 2) $\frac{a^{n+1} - b^{n+1}}{b-a}$ 3) $\frac{b^{n+1} - a^{n+1}}{b-a}$ 4) $\frac{b^n - a^n}{b-a}$
16. The coefficient of x^{24} in the expansion of $(1+x^2)^{12}(1+x^{12})(1+x^{24})$ is
 1) $12C_6$ 2) $12C_6 + 2$ 3) $12C_6 + 4$ 4) $12C_6 + 6$
17. If $0 < x < 1$; then first negative term in the expansion of $(1+x)^{27/5}$ is
 1) 7th term 2) 5th term 3) 8th term 4) 6th term

Approximations :

18. If x is so small that x^2 and higher powers of x are neglected then $\frac{\sqrt{1+x} + \sqrt[3]{1+4x}}{(1+x^2) \sqrt[3]{(1-3x)^2}} =$
 1) $1 + \frac{11x}{12}$ 2) $2 + \frac{35x}{6}$ 3) $1 - \frac{5x}{12}$ 4) $1 + \frac{5x}{12}$
19. If 'c' is small in comparison with l then $\left(\frac{l}{l+c}\right)^{1/2} + \left(\frac{l}{l-c}\right)^{1/2} =$
 1) $2 + \frac{3c}{4l}$ 2) $2 + \frac{3c^2}{4l^2}$ 3) $l + \frac{3c^2}{4l^2}$ 4) $l + \frac{3c}{4l}$
20. If p is nearly equal to q and $n > 1$ such that $\frac{(n+1)p + (n-1)q}{(n-1)p + (n+1)q} = \left(\frac{p}{q}\right)^k$ then the value of k is
 1) n 2) $1/n$ 3) $n+1$ 4) $1/n+1$

21. If x is numerically so small so that x^2 and higher powers of x can be neglected, then $\left(1 + \frac{2x}{3}\right)^{\frac{-1}{2}} (32 + 5x)^{\frac{-1}{5}}$ is approximately equal to:

- 1) $\frac{32+31x}{64}$ 2) $\frac{32+32x}{64}$ 3) $\frac{31+32x}{64}$ 4) $\frac{1-2x}{64}$

22. If x is nearly equal to 1 then value of $\frac{mx^m - nx^n}{m-n}$ is nearly equal to

- 1) $x^m + n$ 2) x^{m-n} 3) $\frac{1}{1-x}$ 4) $\frac{1}{1+x}$

23. If x is nearly equal to 1 then value of $\frac{ax^b - bx^a}{x^b - x^a}$ is nearly equal to

- 1) $\frac{1}{1+x}$ 2) $\frac{1}{1-x}$ 3) $\frac{2}{1+x}$ 4) $\frac{2}{1-x}$

Summation of Infinite Series :

24. The sum of the series $\frac{3}{4 \cdot 8} + \frac{3 \cdot 5}{4 \cdot 8 \cdot 12} + \frac{3 \cdot 5 \cdot 7}{4 \cdot 8 \cdot 12 \cdot 16} + \dots$

- 1) $\sqrt{\frac{3}{2}} - \frac{3}{4}$ 2) $\sqrt{\frac{2}{3}} - \frac{3}{4}$ 3) $\sqrt{\frac{3}{2}} - \frac{1}{4}$ 4) $\sqrt{\frac{2}{3}} - \frac{1}{4}$

25. $\frac{3}{6} + \frac{3 \cdot 5}{6 \cdot 9} + \frac{3 \cdot 5 \cdot 7}{6 \cdot 9 \cdot 12} + \dots \infty =$

- 1) $2\sqrt{3} - 4$ 2) $3\sqrt{3} - 2$ 3) $3\sqrt{3} - 4$ 4) $2\sqrt{3} + 4$

26. $\frac{5}{9 \cdot 18} + \frac{5 \cdot 8}{9 \cdot 18 \cdot 27} + \frac{5 \cdot 8 \cdot 11}{9 \cdot 18 \cdot 27 \cdot 36} + \dots \infty =$

- 1) $\frac{1}{2} \sqrt[3]{\frac{9}{4}} - \frac{11}{18}$ 2) $\frac{3\sqrt[3]{18} - 22}{12}$ 3) $\frac{3\sqrt[3]{9} - 11}{12}$ 4) $\frac{\sqrt[3]{10} - 5}{6}$

27. $1 + \frac{n}{2} + \frac{n(n-1)}{2 \cdot 4} + \frac{n(n-1)(n-2)}{2 \cdot 4 \cdot 6} + \dots + \infty =$

- 1) $1 + \frac{n}{3} + \frac{n(n-1)}{3 \cdot 6} + \frac{n(n+1)(n+1)}{3 \cdot 6 \cdot 9} + \dots$ 2) $1 + \frac{n}{3} + \frac{n(n+2)}{3 \cdot 6} + \frac{n(n+1)(n+1)}{3 \cdot 6 \cdot 9} + \dots$
 3) $1 + \frac{n}{3} + \frac{n(n+1)}{3 \cdot 6} + \frac{n(n+1)(n+2)}{3 \cdot 6 \cdot 9} + \dots$ 4) $1 + \frac{n}{3} + \frac{n(n+2)}{3 \cdot 6} + \frac{n(n+1)(n+2)}{3 \cdot 6 \cdot 9} + \dots$

Numerical value type questions

28. If a_1, a_2, a_3 , are the last three digits of 17^{256} respectively then the value of $4a_1 - 2a_2 - a_3$ is equal to

29. Coefficient of x^{2009} in $(1+x+x^2+x^3+x^4)^{1001}(1-x)^{1002}$ is

30. Given $(1-2x+5x^2+10x^3)(1+x)^n = 1 + a_1x + a_2x^2 + \dots$ and that $a_1^2 = 2a_2$ then the value of n is

31. If $n \in \mathbb{N}$ and $C_k = {}^nC_k$, and $\sum_{k=1}^n k^3 \left(\frac{{}^nC_k}{{}^nC_{k-1}} \right)^2 = \frac{n(n+1)^2(n+2)}{3p}$ then p is

KEY SHEET (LECTURE SHEET)

EXERCISE- I

LEVEL-I

- 1) 2 2) 1 3) 4 4) 1 5) 3 6) 2 7) 1 8) 3
 9) 1 10) 3 11) 1 12) 3 13) 4 14) 3 15) 2 16) 3
 17) 3 18) 1 19) 3 20) 3 21) 4 22) 2 23) 4 24) 2
 25) 1 26) 3 27) 3 28) 3 29) 1 30) 3 31) 4 32) 2
 33) 2 34) 1 35) 1 36) 2 37) 1 38) 2 39) 2 40) 1
 41) 3 42) 4 43) 6 44) 5

LEVEL-II

- 1) d 2) b 3) a 4) a 5) a 6) d 7) b 8) a
 9) c 10) c 11) 3 12) b 13) b 14) c 15) d 16) c
 17) b 18) d 19) bc 20) ab 21) acd 22) ad 23) a 24) c
 25) c 26) b 27) b 28) a 29) A-p; B-q; C-r; D-s
 30) A-q; B-s; C-p; D-r 31) 0 32) 1 33) 0 34) 5 35) 7
 36) 2

EXERCISE-II

LEVEL-I

- 1) 1 2) 1 3) 1 4) 1 5) 1 6) 2 7) 2 8) 2
 9) 2 10) 3 11) 2 12) 1 13) 4 14) 2 15) 3 16) 3
 17) 1 18) 1 19) 1 20) 1 21) 1 22) 1 23) 3 24) 1
 25) 1 26) 2 27) 2 28) 1 29) 3 30) 4 31) 3 32) 3
 33) 2 34) 1 35) 4 36) 1 37) 1 38) 1 39) 1 40) 3

LEVEL-II

- 1) b 2) d 3) c 4) d 5) c 6) c 7) b 8) c
 9) c 10) a 11) b 12) a 13) a 14) a 15) d 16) c
 17) a 18) a 19) ab 20) ad 21) abd 22) ab 23) abc 24) b
 25) d 26) c 27) c 28) a 29) b 30) A-r; B-s; C-p; D-q
 31) A-p; B-r; C-r; D-s 32) 8 33) 9 34) 1 35) 0 36) 0
 37) 9 38) 7

EXERCISE-III

LEVEL-I

- 1) 1 2) 1 3) 4 4) 4 5) 3 6) 2 7) 3 8) 4
 9) 1 10) 3 11) 1 12) 3 13) 2 14) 4 15) 3 16) 2
 17) 3 18) 2 19) 2 20) 2 21) 1 22) 1 23) 2 24) 2
 25) 3 26) 1 27) 3 28) 7 29) 0 30) 6 31) 4

BINOMIAL THEOREM

LECTURE SHEET

EXERCISE-I

LEVEL-I (MAIN)

Single Answer type questions

1. $T_3 = 1 \Rightarrow 5C_2 \left(\frac{1}{x}\right)^3 (x \log_{10} x)^2 = 1 \Rightarrow x = 10$
2. $T_5 + T_6 = 0 \Rightarrow {}^nC_4 a^{n-4} b^4 = {}^nC_5 a^{n-5} b^5 \Rightarrow \frac{a}{b} = \frac{n-4}{5}$
3. $\frac{T_7}{T_{n-5}} = \frac{1}{6} \Rightarrow n = 9$
4. $(x+a)^n = T_1 + T_2 + \dots + T_{n+1}$
 $P = T_1 + T_3 + T_5 + \dots$; $Q = T_2 + T_4 + T_6 + \dots$
 $(p+q)^2 = (x+a)^{2n}$; $(p-q)^2 = (x-a)^{2n}$
 $(p+q)^2 - (p-q)^2 = 4pq = (x+a)^{2n} - (x-a)^{2n}$
5. $(p+q)^2 + (p-q)^2 = 2(p^2+q^2)$
 $\Rightarrow p^2 + q^2 = \frac{(x+a)^{2n} + (x-a)^{2n}}{2}$
6. $(x+a)^n = T_0 + T_1 + T_2 + \dots + T_n$
 $(x+ai)^n = T_0 + i T_1 - T_2 - i T_3 + \dots$
 $= (T_0 - T_2 + T_4 - \dots) + i(T_1 - T_3 + \dots)$
 taking Mod on both sides
7. $2({}^5C_0 x^5 + {}^5C_2 x^3(x^3-1) + {}^5C_4 x(x^3-1)^2)$ this is a 7th degree polynomial
8. ${}^{18}C_{2r+3} = {}^{18}C_{r-3} \Rightarrow r = 6$
9. Coefficient of x^k in $\left(ax^p + \frac{b}{x^2}\right)^n$ is ${}^nC_r a^{n-r} b^r$
 where $r = \frac{np-k}{p+q}$
10. Sum of the coefficients of x^5 and x^{10} is zero $\Rightarrow n = 15$
11. $r = \frac{np}{p+q}$ where $p = 2, q = 1 \Rightarrow n = 6, r = 4$
12. $\left(x - \frac{1}{x}\right)\left(x^2 - \frac{1}{x^2}\right)^3 = \left(x - \frac{1}{x}\right)\left(x^6 - 3x^2 + \frac{3}{x^2} - \frac{1}{x^6}\right)$
 $\Rightarrow$ term independent of x is 0
13. $(1+x)^n \left(1 + \frac{1}{x}\right)^n = \frac{(1+x)^{2n}}{x^n}$ term independent of x
 is coefficient of x^r in $(1+x)^{2n}$ i.e. ${}^{2n}C_n$
14. $\sum_{r=0}^{100} {}^{100}C_r (x-3)^{100-r} \cdot 2^r = (x-1)^{100} = (1-x)^{100}$
 coefficient of x^{53} is ${}^{100}C_{53} (-1)$
15. $(-1)^n \cdot {}^nC_n + (-1)^{n-1} C_{n-1} = -144$

16. $G.E = \frac{(1+x)^{31} - (1+x)^{21}}{6}$
 Coefficient of x^5 is ${}^{31}C_6 - {}^{21}C_6$
17. T_{14} in Given expansion
18. $G.E = (1-x)^{6n}$ Middle term is ${}^{6n}C_{3n} (-x)^{3n}$
19. Coefficients of middle terms in both expansions are same $\Rightarrow 6\alpha^2 = -20\alpha^3 \Rightarrow \alpha = \frac{-3}{10}$
20. $p = 9$ Numerically Greatest terms are T_{10}, T_9
21. $\left(2x - \frac{1}{3x}\right)^{10} = (2x)^{10} \left(1 - \frac{1}{6x^2}\right)^{10}$
 when $x = 1, \frac{(n+1)|x|}{|x|+1} = \frac{11}{7}$; T_2 is N.G.T
22. T_{n+1} in a N.G.T $\Rightarrow T_n < T_{n+1} > T_{n+2}$
 $\Rightarrow {}^{2n}C_{n-1} x^{n-1} < {}^{2n}C_n x^n > {}^{2n}C_{n+1} x^{n+1}$
 $\Rightarrow \frac{n}{n+1} < x < \frac{n+1}{x}$
23. When $x = y = 1, P = \frac{(n+1)|x|}{|x|+1} = \frac{13}{5}$; T_3 is a N.G.T
24. An odd number
25. An even number
26. $(6\sqrt{6}+14)^{2n+1} (6\sqrt{6}-14)^{2n+1} = 20^{2n+1}$
27. $(\sqrt{2}+1)^6 = 197.98$
28. $(101)^{100} - 1 = (1+100)^{100} - 1$
 $= {}^{100}C_1 \cdot 100 + {}^{100}C_2 \cdot 100^2 + \dots + {}^{100}C_{100} \cdot 100^{100}$
 $= 100^2 N$
 where N is integer
29. $8^{2n} - 6 \cdot 2^{2n+1} = (1+63)^n + (1-63)^{2n+1} = 2 + 63k$
30. $(x^2-x-2)^5 = (x-2)^5(x+1)^5$
 $= (x^5 - 5C_1 x^4(2) + 5C_2 x^3 2^2 - 5C_3 x^2 2^3 + 5C_4 x 2^4 - 2^5)(x^5 + 5C_1 x^4 + 5C_2 x^3 + 5C_3 x^2 + 5C_4 x + 5C_5)$
 coefficient of x^5 is -81
31. $(2x-3y+4z)^9 = \sum_{n_1+n_2+n_3=9} \frac{9!}{n_1! n_2! n_3!} (2x)^{n_1} (-3y)^{n_2} (4z)^{n_3}$
32. Number of terms in the expansion of $(x+a)^n + (x-a)^n$ is $\frac{n+1}{2}$, if n is odd integer.
33. Number of terms in $(1+x)^{18}$ is 19
34. Number of terms in $x_1 + x_2 + \dots + x_r = n$ is $(n+r-1)C_{r-1}$

35. Number of terms in $(x+a)^n + (x-a)^n + (x+ai)^n + (x-ai)^n$ is

$$\left[\frac{n+4}{4} \right] \text{ where } [.] \text{ is G.I.F}$$

36. $1_n = (3^{1/3} + 7^{1/3})^{36}$

$$T_{r+1} = {}^{36}C_r 3^{\frac{36-r}{3}} \cdot 7^{r/3} \text{ term is rational if } r = 6, 21, 36$$

$\therefore$ No. of rational terms is 3

$\Rightarrow$ No. of irrational terms is 34

37. $10C_0 2^5 + {}^{10}C_{10} 3^2 = 41$

38. $x+y=1, \sum_{r=0}^n r \cdot {}^nC_r x^r y^{n-r} = \sum_{r=0}^n r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1} x^r y^{n-r}$
 $= nx \sum_{r=1}^n {}^{(n-1)}C_{r-1} x^{r-1} y^{(n-1)-(r-1)} = nx(x+y)^{n-1} = nx$

39. ${}^{14}C_{r-1}, {}^{14}C_r, {}^{14}C_{r+1}$ are in AP

$$(n-2r)^2 = n+2, n=14$$

$$(14-2r)^2 = 16 \Rightarrow 14-2r = \pm 4 \Rightarrow r = 5, 9$$

40. $\frac{a_1}{a_1+a_2} = \frac{{}^nC_1}{{}^nC_1+{}^nC_2} = \frac{2}{n+1};$

$$\frac{a_2}{a_2+a_3} = \frac{3}{n+1}, \frac{a_3}{a_3+a_4} = \frac{4}{n+1} \text{ they are in AP}$$

41. $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \dots$

$$a_1 = n-m = 10; a_2 = {}^nC_2 + {}^mC_2 - mn = 10$$

$$\Rightarrow n = 45, m = 35$$

Numerical value type questions

42. We have, $32 = 2^5; 32^{32} = (2^5)^{32} = 2^{160}$

$$= (3-1)^{160} = {}^{160}C_0 (3)^{160} - {}^{160}C_1 (3)^{159} + \dots$$

$${}^{160}C_{159} (3) + 1$$

$$3(3^{159} - {}^{160}C_1 (3)^{158} + \dots - {}^{160}C_{159}) + 1$$

$$= 3m+1, m \in I^+$$

$$\text{Now, } 32^{32^{1/2}} = 32^{3m+1} = 2^{5(3m+1)} = 2^{15m+5}$$

$$= 2^2 \cdot 2^{5(3m+1)} = 4(8)^{3m+1} = 4(7+1)^{3m+1}$$

$$= 4[({}^{3m+1}C_0 (7)^{3m+1} + {}^{3m+1}C_1 (7)^{3m} + {}^{3m+1}C_2 (7)^{3m-1} + \dots + {}^{3m+1}C_{3m} (7)^{1} + 1]$$

$$= 4[7({}^{3m+1}C_0 (7)^{3m} + {}^{3m+1}C_1 (7)^{3m-1} + \dots + {}^{3m+1}C_{3m} (7)^1 + 1]$$

$$= 4[7k+1], \text{ where } k \text{ is +ve integer} = 28k+4$$

$$\therefore \frac{32^{32^{1/2}}}{7} = 4k + \frac{4}{7}$$

Hence, remainder is 4

43. $\sum_{i=0}^m \binom{10}{i} \binom{20}{m-i} = \sum_{i=0}^m {}^{10}C_i \cdot {}^{20}C_{m-i}$

$${}^{10}C_0 \cdot {}^{20}C_m + {}^{10}C_1 \cdot {}^{20}C_{m-1} + {}^{10}C_2 \cdot {}^{20}C_{m-2} + \dots + {}^{10}C_m \cdot {}^{20}C_0$$

= Coefficient of x^m in expansion of product $(1+x)^{10} \cdot (1+x)^{20}$.

= Coefficient of x^m in the expansion of $(1+x)^{30} = {}^{30}C_m$

To get maximum value of given sum, ${}^{30}C_m$ should be maximum. Which is so, when $m = \frac{30}{2} = 15$

The sum of the digits of m is equal to $1+5 = 6$

44. Here, $f(x) = \sum_{r=1}^n \{r^2 ({}^nC_r - {}^nC_{r-1}) + (2r+1) {}^nC_r\}$

$$\sum_{r=1}^n (r^2 + 2r+1) {}^nC_r - r^2 \cdot {}^nC_{r-1}$$

$$= \sum_{r=1}^n ((r+1)^2 \cdot {}^nC_r - r^2 \cdot {}^nC_{r-1})$$

$$= (n+1)^2 \cdot {}^nC_n - 1^2 \cdot {}^nC_0 = (n+1)^2 - 1 = (n^2 + 2n)$$

$$\therefore f(30) = (30)^2 + 2(30) = 960$$

$$= 30 \times 32 = 30 \times (2)^5 = 30 \times (2)^6$$

Hence $x = 5$

LEVEL-II (ADVANCED)

Single Answer type questions

1. ${}^6C_3 \left(\frac{1}{x^{2(1+\log_{10} x)}} \right)^3 x^{1/4} = 200 \Rightarrow x = 10$

2. $r = \frac{np}{p+q} \Rightarrow r = 6; T_7 = {}^{14}C_6 p^8 q^6$

its maximum value in ${}^{14}C_6$ since $p^4 + q^3 = 2$

3. $t = 4; T_{r+1} = {}^{16}C_r m^{16-r} \left(\frac{b}{m^3} \right)^r$

$$= {}^{16}C_r m^{16-r} \cdot m^{-3r} \cdot b^r = {}^{16}C_r m^{16-4r} \cdot b^r$$

$$16-4r = -12 \Rightarrow 4r = 28 \Rightarrow r = 7$$

Coefficient is ${}^{16}C_7 b^7$

4. $S = (1+x)^{501} - x^{501};$

Hence, coefficient of x^{301} in S is ${}^{501}C_{301}$

5. The highest power of x is 78, to get coefficient of x^{70} , we have to omit the factors containing x^8

- 1) Product of the constant terms of $(x-1)(x^7-7) = 7$
- 2) Product of the constant terms of $(x^2-2)(x^6-6) = 12$
- 3) Product of the constant terms of $(x^3-3)(x^5-5) = 15$
- 4) Product of the constant terms of

$$(x-1)(x^2-2)(x^5-5) = -10$$

- 5) Product of the constant terms of

$$(x-1)(x^3-3)(x^4-4) = -12$$

∴ Required coefficient = 4

6. $G.E = (1+x+x^2+x^3+x^4)^{1001}(1-x)^{1002}(1+7x^{14})$
 $= (1-x^5)^{1001}(1-x)^{-1001}(1-x)^{1002}(1+7x^{14})$
 $= (1-x^5)^{1001}(1-x)(1+7x^{14})$
 $= \sum_{r=0}^{1001} (-1)^r (x)^{5r} \cdot 1001 C_r [(1-x+7x^{14}-7x^{15})]$

Put $r = 403$

Coefficient of x^{2016} is $-1001 C_{403} (-1) = 1001 C_{403}$

7. The coefficient of x^6 in the given expression = coefficient of x^6 in

$$(1+{}^6C_1x^6)(1+{}^5C_1x^5)(1+{}^4C_1x^4)(1+{}^3C_1x^3+{}^3C_2x^6)$$

$$(1+{}^2C_1x^2+{}^2C_2x^4)(1+x)$$

$$= \text{coefficient of } x^6 \text{ in}$$

$$(1+6x^6+5x^5+4x^4)(1+2x^2+3x^3+x^4+6x^5+3x^6)(1+x) = 28$$

8. $(a+b+c+d)^8$

Sum of the terms not containing $c = 3^8$

Sum of the terms not containing $b, c = 2^8$

Sum of the terms containing b not $C = 3^8 - 2^8 = 6305$

9. To get common terms in both the expansions

$$x^{r_1}(y^2)^{13-r_1} = y^{14-r_2}(x^2)^{r_2}$$

$$\therefore r_1 = 2r_2, 26-2r_1 = 14-r_2 \Rightarrow r_1 = 8, r_2 = 4$$

∴ only one term is common

10. n is $(a+b+c)^n + (a+b-c)^n$

$$\Rightarrow a+b=c, (x+c)^n + (x-c)^n \text{ contains } \frac{n}{2} + 1 \text{ terms}$$

$$2[(a+b)^n + {}^nC_2(a+b)^{n-2}c^2 + \dots + c^n],$$

$$2[(n+1)\text{terms} + (n-1)\text{terms} + \dots + 1]$$

$$= \left(\frac{n}{2} + 1\right) \left(\frac{1+n+1}{2}\right) = \frac{(n+2)(n+2)}{2 \times 2} = \frac{(n+2)^2}{4}$$

11. Coeff of x^4 in $\left(1+2x+\frac{3}{x^2}\right)^6$

$$\Rightarrow \frac{6!}{p!q!r!} 1^p 2^q \left(\frac{3}{x^2}\right)^r$$

$$\Rightarrow p+q+r=6 \Rightarrow p=2, q=4, r=0$$

$$= \frac{6!}{2!4!} 2^4 = 240$$

12. $\left(x^3 + \frac{1}{x^3} + 1\right)^{100}$
 $= 1 + {}^{100}C_1 \left(x^3 + \frac{1}{x^3}\right) + {}^{100}C_2 \left(x^3 + \frac{1}{x^3}\right)^2 + \dots + {}^{100}C_{100} \left(x^3 + \frac{1}{x^3}\right)^{100}$
 no terms = $1+2+2+\dots$ (100 times) = 201

13. $(ab+bc+ca)^{10} = (abc)^{10} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)^{10}$
 $\Rightarrow$ Coeff of $a^{10}b^7c^3(ab+bc+ca)^{10}$
 $=$ coeff of $a^0b^{-3}c^{-7}$ in $\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)^{10} = \frac{10!}{3!7!} = 120$

14. $x = (2+\sqrt{3})^n, n \in N, f = x - [x] \frac{f^2}{1-f}$
 $\Rightarrow [x] + f = (2+\sqrt{3})^n$

let $F = (2-\sqrt{3})^n =$ even inter

$[x] + f =$ integer $0 < f < 1$

$$F + f = 1 \quad 0 < f < 1$$

$$0 < f + f < 2$$

$$[x] = f + F = 1$$

$$[[x] + f]F = (2+\sqrt{3})^n (2-\sqrt{3})^n = 1$$

$$\Rightarrow [x] + f = \frac{1}{F} = \frac{1}{1-f}$$

$$\Rightarrow [x] = \frac{1}{1-f} - f = \frac{1-f+f^2}{1-f}$$

$$\Rightarrow \frac{f^2}{1-f} = [x] - 1 = \text{even enter}$$

15. $x > 0$ is an odd integer $x = (\sqrt{2}+1)^n$

$$\Rightarrow f = x - [x]$$

$$\Rightarrow \frac{1-f^2}{f} \Rightarrow x-F = (\sqrt{2}+1)^n (\sqrt{2}-1)^n$$

= even inter $0 < f < 1$

$$[x] + f - F = \text{even inter } 0 < F < 1$$

$$\Rightarrow f - F = \text{Inter } -1 < f - F < 1$$

$$\Rightarrow [x] = \text{even inter } f - F = 0$$

$$\Rightarrow ([x] + f)F = 1 \Rightarrow [x] + f = \frac{1}{f}$$

$$\Rightarrow [x] = \frac{1}{f} - f = \frac{1-f^2}{f} = \text{even integer}$$

16. For T_6 to be numerically greatest term

- $5 < \frac{n+1}{1 + \frac{3/2}{3/3}} < 6 \Rightarrow \frac{23}{2} < n < 14 \Rightarrow n = 12, 13$
 17. $(3-2x)^{15}$ at $x = \frac{4}{3}$; T_8 is numerically greatest term
 and $T_8 = {}^{15}C_7 3^8 \left(\frac{8}{3}\right)^7$
 $\Rightarrow T_7, T_9$ are positive terms and $T_9 - T_7 > 0 \Rightarrow T_9 > T_7$
 $T_7 - T_{11} > 0$ Thus $T_9 > T_7 > T_{11}$
 $\therefore T_9$ is the second largest term.

18. $27^{10} + 7^{51} = (30-3)^{10} + (10-3)^{51} = 3^{10} - 3^{51} + 10\lambda$
 $= (10-1)^5 - 3(10-1)^{25} + 10\lambda$
 $= -1 + 3 + 10\lambda_1 = 2 + 10\lambda_1$
 Remainder = 2

More than one correct answer type questions

19. $T_9 = {}^{10}C_8 \left(\frac{10}{x^{5t}}\right) \cdot x^{8+1}t = 450$
 $\Rightarrow$ where $t = \log_{10}^1$
 $\Rightarrow = 8 + \frac{4}{t} - 5t = x^0$
 $8 + \frac{4}{t} - 5t = 0, (1-2)(5t+2) = 0, t = 2, -\frac{2}{5}$
 $\Rightarrow x = 10^2 \Rightarrow 10^{-25}$
20. a, b, c, d are any four consecutive coefficients of $(1+x)^n$
 then which of the following is are correct
 A) $a = {}^nC_1, b = {}^nC_2, c = {}^nC_3, d = {}^nC_4$

$$\frac{a}{a+b} + \frac{c}{c+d} = \frac{1}{1 + \frac{{}^nC_2}{{}^nC_1}} + \frac{1}{1 + \frac{{}^nC_4}{{}^nC_3}}$$

$$= \frac{2}{n+1} + \frac{4}{n+1} = \frac{6}{n+1}$$

$$\Rightarrow \frac{2b}{b+c} = \frac{2}{1 + \frac{{}^nC_3}{{}^nC_2}} = \frac{6}{n+1}$$

$$\Rightarrow \frac{a}{a+b}, \frac{b}{b+c}, \frac{c}{c+d} \text{ are in AP. Am} > \text{Gm} :$$

$$\frac{b}{b+c} > \left(\frac{9}{a+b} \times \frac{c}{c+d}\right)^{1/2}$$
21. $P_{(1)}(x) = 4 + a_1x + a_2x^2 + \dots$ constant term in $P_1(x)$ is 4

$\Rightarrow$ Constant term in $P_1(x)$ is 4

Coef of x in $P_2(x)$ is $-4^2 \Rightarrow$ coef of x in P_4 is -4^4

$\Rightarrow$ Coef of x^2 in $P_2(x)$ is 20 $\Rightarrow$ ie $\frac{4^{24-1} - 4^{4-1}}{3}$

22. Since a and b rational,

we have $(3-a\sqrt{2})^{100} + (3-b\sqrt{2})^{100} = 7-5\sqrt{2}$

R.H.S < 0 but L.H.S > 0

$\therefore$ there are no such pair (a, b)

b) No. of different terms $= 2n+1 = 401$

$$c) T_{r+1} = {}^{16}C_r \left(\frac{x}{\cos\theta}\right)^{16-r} \left(\frac{1}{x\sin\theta}\right)^r$$

$$= {}^{16}C_r \frac{1}{(\cos\theta)^{16-r} (\sin\theta)^r} x^{16-2r}$$

Term is independent $16-2r = 0 \Rightarrow r = 8$

$$\text{Independent term } T_9 = {}^{16}C_8 \frac{2^8}{\sin^8 20}$$

$$t_1 = {}^{16}C_8 \cdot 2^8; t_2 = {}^{16}C_8 \cdot 2^{12}; \frac{t_1}{t_2} = 2^4 = 16$$

$$d) \text{ sum of the roots} = {}^{2001}C_2 \cdot \frac{1}{4} \cdot \frac{2}{2001C_1} = 500.$$

Linked comprehension type questions

Passage-I:

23. $f(x) = 3^{2x} + 3^x + 1 \Rightarrow f(x+3) = 3^{2x+6} + 3^{x+3} + 1$
 $\Rightarrow f(x+3) - 3^6 f(x) = -702 \times 3^6 - 728$
 24. $f(100)$ is divisible by 13
 $\Rightarrow f(1001)$ is divisible by 13
 $\Rightarrow f(2007)$ is not divisible by 13
 25. $f(10)$ is divisible by 13
 $\Rightarrow f(51)$ leaves remained 3 when divisible by 13

Passage-II:

26. $E = \frac{1}{x}(x+2)^n - (x+1)^n \quad E = \frac{1}{x}(x+2)^{n+1} - 2^{n+1}$
 27. $E = \frac{1}{2}[2x+1]^n - (x+1)^n$
 28. $E = \frac{1}{x}(x+2)^{n+1} - 2^{n+1}$

Matrix matching type questions

29. A) $5^{97} = 5 \cdot 5^{96} = 5 \cdot (25)^{48} = 5 \cdot (26-1)^{48}$
 $= 5[(26)^{48} - 48(26)^{47} + \dots + {}^{48}C_{47}(26) + 1]$
 $= 5(52\lambda) + 5 \text{ remainder } 5$

$$B) 53^{53} - 33^3 = (53^{53} - 3^{53}) + 3^{53} - [(33^3 - 3^3) + 3^3]$$

$$\Rightarrow 53^{53} - 3^{53} \text{ is divisible by } 50$$

Then it is divisible by 10

$$\Rightarrow 33^3 - 3^3 \text{ is divisible by } 30 \text{ Hence by } 10$$

$$\Rightarrow 3^{53} - 3^3 = 3^3(3^{50} - 1)$$

$$\Rightarrow = 3^3[(10-1)^{25} - 1] = 23^3[10K-2] \Rightarrow = 270K-54$$

$$\Rightarrow = 270K-60+6 \Rightarrow = 10\lambda + 6 \text{ remainder } 6$$

$$C) 27^{10} + 7^{51} = (27^{10} - 7^{10}) + 7^{10} + 7^{51} \text{ divisible by } 20$$

$$\text{Hence by } 10 \quad 7^{10} + 7^{51} = (7^2)^5 + (50-1)^{25} \cdot 7$$

$$= (50-1)^5 + (50-1)^{25} \cdot 7$$

$$\Rightarrow 10\lambda - 1 + 7(10k-1) \Rightarrow = 10\mu - 8 \text{ remainder } 2$$

$$D) 13^{99} - 19^{93} = (1+12)^{99} - (1+18)^{93}$$

$$= 99 \times 12 - 93 \times 18 \text{ is divisible by } 81$$

$$\text{also } 13^{99} - 19^{93} \text{ is divisible by } 162$$

$$30. (x+y-z+w)^6$$

$$A) \text{ no of terms } = {}^{n+r-1}C_{r-1} = {}^9C_3 = 84;$$

$$b) 7^{2011} = 7[50-1]^{1005}$$

$$= 7[{}^{1005}C_0(50)^{1005} - {}^{1005}C_1(50)^{1004} + \dots + {}^{1005}C_{1004}50 - 1]$$

$$2011 \text{ is two remainder is } = 43$$

$$C) (3\sqrt{7} + 5\sqrt{11})^{1001} = \left[\frac{1001}{15} \right] = 66$$

$$D) 2^{2011} \cdot 127 = 2^2 (2^7)^{287} \Rightarrow = 2^2 [127+1]^{287}$$

Reminder is 4

Integer answer type questions

$$31. f(x) = (\sqrt{3} + \sqrt{2} + 1)^6 + (\sqrt{3} + \sqrt{2} - 1)^6 \\ + (\sqrt{3} - \sqrt{2} + 1)^6 + (-\sqrt{3} + \sqrt{2} + 1)^6$$

Is always an integer $\therefore \{f(x)\} = 0$.

$$32. |m-n| = 1, \text{ where } m=18, n=19$$

$$33. x = (3+2\sqrt{2})^n \Rightarrow x(1-x+[x]) = 1$$

$$y = (9+4\sqrt{5})^n \Rightarrow y(1-y+[y]) = 1$$

$$\Rightarrow (x-y) - (x^2 - y^2) + x[x] - y[y] = 0$$

$$34. x = 5^{5^t - (24 \text{ times})} \Rightarrow = 5^{5^t - (24 \text{ times})} \Rightarrow 5^{2m+1} = 25^m \cdot 5$$

$$= (24+1)^m \cdot 5 \quad m \in N = 5+24K. \text{ Hence the remind is } 5$$

$$35. 5^{81} + \lambda \Rightarrow 5(5^2)^{40} + \lambda \Rightarrow = 5[26-1]^{40} + \lambda$$

$$\Rightarrow = 26K + 5 + \lambda \Rightarrow \lambda = 21 \Rightarrow \frac{\lambda}{3} = 7$$

$$36. S = \left(\frac{n+1}{2} \right) (2a+nd). A = (2a+nd)2^{n-1}$$

$$\Rightarrow \frac{S}{A} = \frac{n+1}{2^n}$$

EXERCISE-II

LEVEL-I (MAIN)

Single Answer type questions

$$1. G.E = (2a+nd)2^{n-1}, \text{ where } a=1, d=3$$

$$2. \sum_{r=0}^n (-1)^r (3+4r) \cdot C_r = 0$$

$$3. \left(1 + \frac{C_0}{C_1}\right) \left(1 + \frac{C_1}{C_2}\right) \left(1 + \frac{C_2}{C_3}\right) \dots \left(1 + \frac{C_{n-1}}{C_n}\right) = \frac{(n+1)^n}{n!}$$

$$4. {}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r, \text{ Numerator} = {}^{n+m}C_m; \text{ Denominator} = {}^{m+n}C_n$$

$$5. \sum_{r=0}^{n-1} \frac{C_r}{C_r + C_{r+1}} = \sum_{r=0}^{n-1} \frac{1}{1 + \frac{C_{r+1}}{C_r}} = \sum_{r=0}^{n-1} \frac{r+1}{n+1} = \frac{n}{2}$$

$$6. \sum_{r=0}^{16} {}^{16}C_r = 2^{16}$$

$${}^{16}C_0 + {}^{16}C_1 + \dots + {}^{16}C_7 = {}^{16}C_9 + \dots + {}^{16}C_{16}$$

$$= K, \text{ then } 2K + {}^{16}C_8 = 2^{16} \Rightarrow K = 2^{15} - \frac{1}{2} {}^{16}C_8$$

$$7. \sum_{r=0}^{2n+1} (-1)^r (2n+1) C_r = 0$$

$$8. \sum_{r=0}^n (a+rd) \cdot {}^nC_r = (2a+nd)2^{n-1}$$

$$9. (1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$$

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + nC_n(-x)^{n-1}$$

$$\text{put } x = -1, C_1 - 2C_2 + 3C_3 - \dots + nC_n(-1)^{n-1} = 0$$

$$10. \text{ Differentiating the expansion } (1+x)^n \text{ two times and put } x = 1$$

$$11. \text{ formula } \frac{2^{n+1}-1}{n+1} = \frac{C_0}{1} + \frac{C_1}{2} + \dots + \frac{C_n}{n+1}$$

$$12. \frac{C_1}{2} + \frac{C_3}{4} + \dots = \frac{2^n-1}{n+1}$$

$$13. \text{ Integrating the expansion } x(1+x)^n \text{ with the limits } 0 \text{ to } 1$$

$$14. \text{ We know that } C_0x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + \dots + \dots +$$

$$+ C_n\frac{x^{n+1}}{n+1} = \frac{(1+x)^{n+1}-1}{n+1}$$

Integrating this expansion with limits 0 to 1

$$15. S = 0.C_0^2 + 1.C_1^2 + 2.C_2^2 + \dots + n.C_n^2$$

$$S = n.C_0^2 + (n-1).C_1^2 + \dots + 0.C_n^2$$

$$2S = n(C_0^2 + C_1^2 + \dots + C_n^2)$$

$$= n \cdot 2^n C_n \Rightarrow S = \frac{n}{2} (2^n) C_n$$

$$16. (1+x)^{2n} = C_0 + C_1 x + C_2 x^2 + \dots + 2nC_{2n} x^{2n}$$

diff. with respect to x ,

$$2n(1+x)^{2n-1} = C_1 + C_2(2x) + C_3(3x^2) + \dots +$$

$$2nC_{2n} 2n(x)^{2n-1} \rightarrow (1)$$

$$\left(1 - \frac{1}{x}\right)^{2n} = C_0 - \frac{C_1}{x} + \frac{C_2}{x^2} - \frac{C_3}{x^3} + \dots + \frac{C_{2n}}{x^{2n}} \rightarrow (2)$$

Multiplying (1) and (2), equating the coefficients of $\frac{1}{x}$ on both sides of product

$$17. 2nC_{n-r} = C_0 C_r + C_1 C_{r-1} + \dots + C_{r-1} C_1 + C_r C_0$$

$$18. C_0^2 - C_1^2 + C_2^2 + (-1)^n C_n^2 = \begin{cases} 0 & \text{if } n \text{ odd} \\ (-1)^{n/2} n C_{n/2} & \text{if } n \text{ is even} \end{cases}$$

$$19. C_0^2 - C_1^2 + C_2^2 + (-1)^n C_n^2 = \begin{cases} 0 & \text{if } n \text{ odd} \\ (-1)^{n/2} n C_{n/2} & \text{if } n \text{ is even} \end{cases}$$

$$20. {}^{(n+1)}C_0 + {}^nC_1 + {}^{(n-1)}C_2 + \dots + 1.C_n = (n+2).2^{n-1}$$

$$21. {}^{(m+n)}C_r = {}^mC_0 \cdot {}^nC_r + {}^mC_1 \cdot {}^nC_{r-1} + \dots + {}^mC_r \cdot {}^nC_0$$

$$22. a \sum_{r=1}^n (-1)^r n C_r + \sum_{r=1}^n (-1)^r r C_r = a$$

$$23. t_n = \sum_{r=0}^n \frac{n-r}{n C_{n-r}} = n \sum_{r=0}^n \frac{1}{n C_r} - \sum_{r=0}^n \frac{r}{n C_r};$$

$$t_n = n.S_n - t_n \Rightarrow t_n = \frac{n}{2} S_n$$

$$24. \frac{f(1)+f(-1)}{2}$$

$$25. \frac{f(1)-f(-1)}{2}$$

$$26. (1+2\sqrt{x})^{40} = {}^{40}C_0 + {}^{40}C_1 (2\sqrt{x}) + {}^{40}C_2 (2\sqrt{x})^2 +$$

$$\dots + {}^{40}C_{40} (2\sqrt{x})^{40}$$

$$(1-2\sqrt{x})^{40} = {}^{40}C_0 - {}^{40}C_1 (2\sqrt{x}) + {}^{40}C_2 (2\sqrt{x})^2 +$$

$$\dots - {}^{40}C_{40} (2\sqrt{x})^{40}$$

$$2({}^{40}C_0 + {}^{40}C_2 (2\sqrt{x})^2 + \dots + {}^{40}C_{40} (2\sqrt{x})^{40}) =$$

$$(1+2\sqrt{x})^{40} - (1-2\sqrt{x})^{40}$$

$$\text{put } x = 1, \text{ we get required sum} = \frac{3^{40} + 1}{2}$$

$$27. \text{Put } x = y = 1, \text{ we get } n = 9$$

greatest Binomial coefficient is 9C_4 (or) 9C_5

$$28. a_1 + a_2 (2x) + a_3 (3x^2) + \dots + a_{20} (20x^{19}) = 10$$

$$(1+2x+3x^2)^9 (6x+2)$$

$$2a_2 + a_3 (6x) + \dots + a_{20} 380x^{18}$$

$$= 10\{9(1+2x+3x^2)^8 (6x+2) + 6(1+2x+3x^2)^9\}$$

$$\text{put } x = 0 \text{ in both conditions we get } \frac{a_2}{a_1} = \frac{210}{20} = 10.5$$

$$29. (1+x+x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$$

differentiate w.r.to x , and put $x = -1$

$$30. \text{differentiate w.r.to } x \text{ and put } x = 1$$

$$31. \text{Differentiate G.E., and Put } x=1$$

$$32. \frac{f(1)+f(-1)}{2} - 1$$

$$33. \frac{f(1)-f(-1)}{2}$$

$$34. \frac{f(1)+f(-1)}{2}$$

$$35. \text{Put } x = i \text{ and equating real and equal parts}$$

$$36. \frac{f(1)+f(w)+f(w^2)}{3}$$

$$37. \text{Put } x = i \text{ and taking modulus on both sides}$$

$$38. \frac{f(1)+f(-1)+f(i)+f(-i)}{4}$$

$$39. \frac{1}{a-1} \cdot \frac{1}{a-1} \cdot \frac{1}{a-1} = \frac{1}{(a-1)^3}$$

$$40. {}^nC_4 + {}^nC_3 + {}^nC_2 + {}^nC_1$$

LEVEL-II (ADVANCED)

Single Answer type questions

$$1. \text{Let } T_{r+1} \geq T_r \Rightarrow \frac{{}^nC_r}{(r+1)(r+2)(r+3)} \geq \frac{{}^nC_{r-1}}{r(r+1)(r+2)}$$

$$\Rightarrow \frac{n-r+1}{r} \geq \frac{r+3}{r} \Rightarrow n-2 \geq 2r \text{ for } n = 50, r \leq 24$$

So, 24th and 25th terms of sequence are greatest.

$$2. (1+\omega)^n = C_0 + C_1 \omega + C_2 \omega^2 + \dots + C_n \omega^n$$

$$(1+1)^n = C_0 + C_1 + C_2 + \dots + C_n$$

$$(1+\omega)^n (2^n) = 2C_0 + C_1(1+\omega) + C_2(1+\omega^2) + \dots$$

$$= 2^n + (-1)^n \omega^{2n} = 2^n - 1, n \in 3\lambda$$

$$3. \text{Coefficient of } x^{2r} \text{ in } (1+x+x^2)^n = \text{coefficient of } x^{2r} \text{ in}$$

$$\left[\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \right]^n = \text{coefficient of } x^{2r} \text{ in } [1+x(1+x)]^n$$

$$= \sum_{k=0}^r {}^nC_{r-k} {}^{r+k}C_{r-k}$$

$$= \sum_{k=0}^r \frac{n!}{(r+k)!(n-r-k)!} \times \frac{(r+k)!}{(2k)!(r-k)!}$$

$$= \sum_{k=0}^r {}^nC_{2k} {}^{n-2k}C_{r-k}$$

Coefficient in x^2 in

$$\left(\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right)^n = x^{2r} \text{ in } \sum_{k=0}^n {}^nC_k \left(x + \frac{1}{2} \right)^{2k} \left(\frac{3}{4} \right)^{n-k}$$

$$= \sum_{k=0}^n {}^nC_k {}^{r+k}C_{2r} \left(\frac{1}{2} \right)^{2k-2r} \left(\frac{3}{4} \right)^{n-k}$$

$\Rightarrow$ Ration of both expansions = 1

4. We have $\frac{1}{3^n C_r} = \frac{(3n-r)!r!}{(3n)!}$

$$= \frac{3n+1}{3n+2} \left[\frac{[(3n+1-r)+(r+1)]\{(3n-r)!r!\}}{(3n+1)!} \right]$$

$$= \frac{3n+1}{3n+2} \left[\frac{1}{3^{n+1} C_r} + \frac{1}{3^{n+1} C_{r+1}} \right] \quad \text{So, } \sum_{r=1}^{3n-1} \frac{(-1)^{r-1} r}{3^n C_r}$$

$$= \sum_{r=1}^{3n-1} (-1)^{r-1} r \left[\frac{1}{3^{n+1} C_r} + \frac{1}{3^{n+1} C_{r+1}} \right] \left(\frac{3n+1}{3n+2} \right)$$

$$= \left[\left(\frac{1}{C_1} + \frac{1}{C_2} \right) - 2 \left(\frac{1}{C_2} + \frac{1}{C_3} \right) + 3 \left(\frac{1}{C_3} + \frac{1}{C_4} \right) \dots \right] \left(\frac{3n+1}{3n+2} \right)$$

$$= \left(\frac{3n+1}{3n+2} \right) \left[\frac{1}{C_1} - \frac{1}{C_2} + \frac{1}{C_3} - \frac{1}{C_4} \dots - \frac{1}{C_{3n}} + \frac{3n}{C_{3n}} \right]$$

$$= \left(\frac{3n+1}{3n+2} \right) \frac{3n}{(3n+1)} = \frac{3n}{3n+2}$$

5. $\sum_{r=0}^n \frac{(r+2)C_r}{n+1} = \sum_{r=0}^n \left(1 + \frac{1}{r+1} \right) C_r = 2^n + \frac{1}{n+1} \sum_{r=0}^n {}^{n+1}C_{r+1}$

$$= 2^n + \frac{2^{n+1} - 1}{n+1}$$

6. ${}^nC_1 x(1-x)^{n-1} + 2 \cdot {}^nC_2 x^2(1-x)^{n-2}$

$$+ 3 \cdot {}^nC_3 x^3(1-x)^{n-3} + \dots + n \cdot {}^nC_n x^n$$

$$= \sum_{r=1}^n r \cdot {}^nC_r x^r (1-x)^{n-r} = \sum_{r=1}^n r \cdot \frac{n}{r} {}^{n-1}C_{r-1} x^r (1-x)^{n-r}$$

$$= nx[x+(1-x)]^{n-1} = nx = 8x(n=8)$$

7. $\sum_{r=0}^{2n} a_r (x-2)^2 = \sum_{r=0}^{2n} b_r (x-3)^2$ $a_k = 1$ $k \geq n$ $b_n =$

$$\text{Let } x-3=t \quad \sum_{r=0}^{2n} b_r t^r = \sum_{r=0}^{2n} a_r (1+t)^r$$

$$\Rightarrow b_n = \text{Coeff of } t^n \text{ in } (1+t)^n + (1+t)^{n+1} + \dots + (1+t)^{2n}$$

$$\therefore a_r = 1 \quad \Rightarrow b_n = \frac{(1+t)^n [(1+t)^{n+1} - 1]}{1+t-1}$$

$$\Rightarrow b_n = \text{coeff of } x^{n+1} \text{ in } (1+t)^{2n+1} - (1+t)^n$$

$$\Rightarrow {}^{2n+1}C_{n+1} - {}^{2n+1}C_n$$

8. If $n \in N$, then

$$\sum_{r=0}^n (-1)^r \cdot {}^nC_r \cdot \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \dots \text{to } m \text{ terms} \right]$$

$$\text{A) } \sum_{r=0}^n (-1)^r {}^nC_r \left(\frac{1}{2} \right)^2 + \sum_{r=0}^n (-1)^r {}^nC_r \left(\frac{3}{8} \right)^2 + \dots \text{m terms}$$

$$= \left(\frac{1}{2} \right)^n + \left(\frac{1}{4} \right)^n + \left(\frac{1}{8} \right)^n + \dots \text{m terms}$$

$$\Rightarrow = \frac{1}{2^n} \left[\frac{1 - \left(\frac{1}{2^n} \right)^m}{1 - \frac{1}{2^n}} \right] = \frac{2^{mn} - 1}{2mn(2^n - 1)}$$

9. In ΔABC $\sum_{r=0}^n {}^nC_r a^{n-r} b^r \cos(rA - (n-r)B) =$

$$= \sum_{r=0}^n {}^nC_r a^{n-r} b^r e^{i[rA - (n-r)B]}$$

$$= \sum_{r=0}^n {}^nC_r (ae^{-iB})^{n-r} (be^{iA})^r \Rightarrow (ae^{-iB} + be^{iA})^n$$

$$= (ae^{-iB} + be^{iA})^n \Rightarrow [a(\cos B - i \sin B) + b(\cos A + i \sin A)]^n$$

$$= (a \cos B + b \cos A)^n = e^n$$

10. For all $n \in N$, $2^{3^n} + 1$ is of the form $3^2 K$, $K \in Z$

11. If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$ $6(a_0 + a_6 + a_{12} + a_{18} + \dots)$

$$= \text{Put } x = 1, \cos \frac{\pi}{3}, \cos \frac{2\pi}{3}, -1, \cos \frac{4\pi}{3}, \cos \frac{5\pi}{3}$$

and adding $6(a$

$$= \sum_{k=0}^r \frac{n!}{(r+k)!(n-r-k)!} \times \frac{(r+k)!}{(2k)!(r-k)!}$$

$$= \sum_{k=0}^r {}^nC_{2k} {}^{n-2k}C_{r-k}$$

Coefficient in x^2 in

$$\left(\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right)^n = x^{2r} \text{ in } \sum_{k=0}^n {}^nC_k \left(x + \frac{1}{2} \right)^{2k} \left(\frac{3}{4} \right)^{n-k}$$

$$= \sum_{k=0}^n {}^nC_k {}^{r+k}C_{2r} \left(\frac{1}{2} \right)^{2k-2r} \left(\frac{3}{4} \right)^{n-k}$$

$\Rightarrow$ Ration of both expansions = 1

4. We have $\frac{1}{3^n C_r} = \frac{(3n-r)!r!}{(3n)!}$

$$= \frac{3n+1}{3n+2} \left[\frac{[(3n+1-r)+(r+1)] \{ (3n-r)!r! \}}{(3n+1)!} \right]$$

$$= \frac{3n+1}{3n+2} \left[\frac{1}{3^{n+1} C_r} + \frac{1}{3^{n+1} C_{r+1}} \right] \quad \text{So, } \sum_{r=1}^{3n-1} \frac{(-1)^{r-1} r}{3^n C_r}$$

$$= \sum_{r=1}^{3n-1} (-1)^{r-1} r \left[\frac{1}{3^{n+1} C_r} + \frac{1}{3^{n+1} C_{r+1}} \right] \left(\frac{3n+1}{3n+2} \right)$$

$$= \left[\left(\frac{1}{C_1} + \frac{1}{C_2} \right) - 2 \left(\frac{1}{C_2} + \frac{1}{C_3} \right) + 3 \left(\frac{1}{C_3} + \frac{1}{C_4} \right) \dots \right] \left(\frac{3n+1}{3n+2} \right)$$

$$= \left(\frac{3n+1}{3n+2} \right) \left[\frac{1}{C_1} - \frac{1}{C_2} + \frac{1}{C_3} - \frac{1}{C_4} \dots - \frac{1}{C_{3n}} + \frac{3n}{C_{3n}} \right]$$

$$= \left(\frac{3n+1}{3n+2} \right) \frac{3n}{(3n+1)} = \frac{3n}{3n+2}$$

5. $\sum_{r=0}^n \frac{(r+2)C_r}{n+1} = \sum_{r=0}^n \left(1 + \frac{1}{r+1} \right) C_r = 2^n + \frac{1}{n+1} \sum_{r=0}^n {}^{n+1}C_{r+1}$

$$= 2^n + \frac{2^{n+1}-1}{n+1}$$

6. ${}^nC_1 x(1-x)^{n-1} + 2 \cdot {}^nC_2 x^2(1-x)^{n-2}$

$$+ 3 \cdot {}^nC_3 x^3(1-x)^{n-3} + \dots + n \cdot {}^nC_n x^n$$

$$= \sum_{r=1}^n r \cdot {}^nC_r x^r (1-x)^{n-r} = \sum_{r=1}^n r \cdot \frac{n}{r} {}^{n-1}C_{r-1} x^r (1-x)^{n-r}$$

$$= nx[x+(1-x)]^{n-1} = nx = 8x(n=8)$$

7. $\sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r$ $a_k = 1$ $k \geq n$ $b_n =$

$$\text{Let } x-3=t \quad \sum_{r=0}^{2n} b_r t^r = \sum_{r=0}^{2n} a_r (1+t)^r$$

$$\Rightarrow b_n = \text{Coeff of } t^n \text{ in } (1+t)^n + (1+t)^{n+1} + \dots + (1+t)^{2n}$$

$$\therefore a_r = 1 \quad \Rightarrow b_n = \frac{(1+t)^n [(1+t)^{n+1} - 1]}{1+t-1}$$

$$\Rightarrow b_n = \text{coeff of } x^{n+1} \text{ in } (1+t)^{2n+1} - (1+t)^n$$

$$\Rightarrow = {}^{2n+1}C_{n+1} - {}^{2n+1}C_n$$

8. If $n \in N$, then

$$\sum_{r=0}^n (-1)^r \cdot {}^nC_r \cdot \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \dots \text{to } m \text{ terms} \right]$$

$$A) \sum_{r=0}^n (-1)^r {}^nC_r \left(\frac{1}{2} \right)^2 + \sum_{r=0}^n (-1)^r {}^nC_r \left(\frac{3}{4} \right)^2 + \dots m \text{ terms}$$

$$= \left(\frac{1}{2} \right)^n + \left(\frac{1}{4} \right)^n + \left(\frac{1}{8} \right)^n + \dots m \text{ terms}$$

$$\Rightarrow = \frac{1}{2^n} \left[\frac{1 - \left(\frac{1}{2^n} \right)^m}{1 - \frac{1}{2^n}} \right] = \frac{2^{mn} - 1}{2mn(2^n - 1)}$$

9. In a ΔABC $\sum_{r=0}^n {}^nC_r a^{n-r} b^r \cos(rA - (n-r)B) =$

$$= \sum_{r=0}^n {}^nC_r a^{n-r} b^r e^{i[rA - (n-r)B]}$$

$$= \sum_{r=0}^n {}^nC_r (ae^{-iB})^{n-r} (be^{iA})^r \Rightarrow (ae^{-iB} + be^{iA})^n$$

$$= (ae^{-iB} + be^{iA})^n \Rightarrow [a(\cos B - i \sin B) + b(\cos A + i \sin A)]^n$$

$$= (a \cos B + b \cos A)^n = c^n$$

10. For all $n \in N$, $2^{3^n} + 1$ is of the form $3^2 K$, $K \in Z$

11. If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$ $6(a_0 + a_6 + a_{12} + a_{18} + \dots)$

$$= \text{Put } x = 1, \cos \frac{\pi}{3}, \cos \frac{2\pi}{3}, -1, \frac{\cos 4\pi}{3}, \frac{\cos 5\pi}{3}$$

$$\text{and adding } 6(a_0 + a_6 + a_{12} + a_{18} + \dots)$$

$$= 3^n + (1+i\sqrt{3})^n + 1 + (1-i\sqrt{3})^n$$

$$\Rightarrow = 3^n + 1 + 2^{n+1} \cos \frac{n\pi}{3}$$

12. $C_1 + C_3 + C_5 + C_7 + \dots = 2^{n-1}$

$$\Rightarrow C_1 - C_3 + C_5 - C_7 + \dots = \operatorname{Im} (1+i)^n = 2^{n/2} \sin \frac{n\pi}{4}$$

$$C_3 + C_5 + \dots = \frac{2^{n-1} - 2^{n/2} \sin \frac{n\pi}{4}}{2}$$

13. If k and n are positive integers

$$S_k = 1^k + 2^k + 3^k + \dots + n^k \text{ then}$$

$$\sum_{r=1}^m {}^{m+1}C_r (S_r) =$$

$$\sum_{r=1}^n {}^{m+1}C_r S_r = \sum_{r=1}^{11} {}^{m+1}C_r [1^r + 2^r + \dots + n^r]$$

$$\Rightarrow \sum_{k=1}^n \sum_{r=1}^n {}^{m+1}C_r K^2 = \sum_{k=1}^n (1+K)^{m+1} - 1 - k^{m+1}$$

$$\sum_{k=1}^n [(1+K)^{m+1} - K^{m+1}] - \sum_{k=1}^n 1 \Rightarrow (n+1)^{m+1} - 1 - n$$

14. The sum of the coefficients of terms not containing $c=3^a$ (putting $a=b=d=1, c=0$)

The sum of the coefficients of terms not containing both b and $c=2^a$ (putting $a=d=1, b=c=0$) So, the required number $= 3^a - 2^a$

15. $a_0^2 - a_1^2 + a_2^2 - \dots + a_n^2 = \text{coefficient of } x^{2n} \text{ in } (1+x+x^2)^n (x^2-x+1)^n$

16. The value of $\sum_{0 \leq i < j \leq n} i {}^nC_j \sum_{0 \leq i < j \leq n} i {}^iC_j$

$$= \sum_{r=0}^{n-1} {}^nC_r [(r+1) + (r+2) + \dots + n]$$

$$\Rightarrow \sum_{r=0}^{n-1} {}^nC_r \left[\left(\frac{n-r}{2} \right) (r+1) + n \right]$$

17. $\sum_{j=0}^n \sum_{i=0}^n (-1)^{i+j-1} {}^nC_i {}^nC_j = \sum_{i=0}^n (-1)^{2i-1} {}^nC_i + 2s$

$$\Rightarrow 0 = -\sum_{i=0}^n ({}^nC_i)^2 + 2s$$

$$\Rightarrow 2s = \sum_{i=0}^n ({}^nC_i)^2 \Rightarrow s = \frac{1}{2} \sum_{i=0}^n ({}^nC_i)^2 = \sum_{i=0}^{n-1} {}^nC_n = {}^{2n-1}C_n$$

18. We have $S_1 + S_2 + S_3 = \sum_{i=0}^{100} \sum_{j=0}^{100} C_i C_j$

$$= ({}^{100}C_0 + {}^{100}C_1 + \dots + {}^{100}C_{100})^2 = 2^{200} = 16^{50} = a^b$$

Hence least value of $a+b$ is 66

More than one correct answer type questions

19. Given series $S = \sum_{r=0}^n (-1)^r {}^nC_r (2^{n-2r}) C_m$

$$= \sum_{r=0}^n (-1)^r {}^nC_r \cdot \text{coefficient of } x^m \text{ in } (1+x)^{2n-2r}$$

$$= \text{coefficient of } x^m \text{ in } \sum_{r=0}^n {}^nC_r (-1)^r ((1+x^2))^{n-r}$$

$$= \text{coefficient of } x^m \text{ in } ((1+x^2)-1)^n$$

$$= \text{coefficient of } x^{m-n} \text{ in } (x+2)^n$$

$$= \begin{cases} {}^nC_m 2^{2n-m} & \text{if } m \geq n \\ 0 & \text{if } m < n \end{cases}$$

20. ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - \dots - {}^{20}C_{15}$

$$= \text{coefficient of } x^{15} \text{ in } (1-x)^{20} (1+x+x^2+\dots) = -{}^{19}C_{15}$$

$$16. {}^{20}C_0 - 15.{}^{20}C_1 + 14.{}^{20}C_2 - \dots - {}^{20}C_{15}$$

$$= \text{coefficient of } x^{15} \text{ in } (1-x)^{20} (1+2x+3x^2+\dots) = {}^{18}C_{15}$$

21. $(1+\sqrt{2})^n = x_n + y_n \sqrt{2}$ x_n, y_n are integers

$$(1+\sqrt{2})^n = x_n + y_n \sqrt{2} \Rightarrow (1-\sqrt{2})^n = x_n - y_n \sqrt{2}$$

$$\Rightarrow (-1)^n = x_n^2 - 2y_n^2$$

$$x_{n+1} + y_{n+1} \sqrt{2} = (1+\sqrt{2})^{n+1} = (1+\sqrt{2})(1+\sqrt{2})^n$$

$$= (1+\sqrt{2})(x_n + y_n \sqrt{2}) = (x_n + 2y_n) + \sqrt{2}(x_n + y_n)$$

$$\Rightarrow x_{n+1} = x_n + 2y_n, y_{n+1} = x_n + y_n$$

22. If $(x^{2000} + x^{2008} + 2)^{2010} = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$

find then value of

$$a_0 - \frac{1}{2}a_1 - \frac{1}{2}a_2 + a_3 - \frac{1}{2}a_4 - \frac{1}{2}a_5 + a_6 + \dots$$

$$A) x=W, W^2 \Rightarrow 1 = a_0 + a_1 W + a_2 W^2 + a_3 + \dots$$

$$1 = a_0 + a_1 W^2 + a_2 W + a_3 + \dots$$

$$\Rightarrow 2 = 2a_0 - a_1 - a_2 + 2a_3$$

$$+ \dots, a_0 - \frac{1}{2}a_1 - a_2 + a_3 + \dots = 1$$

23. $ac > b^2$ sum of the coefficients in the expansion of $(ax^2x^2 + 2bxx + c)^n$

$$\text{Let } \alpha = 1 (ax^2+2bx+c)^n \text{ Discriminant of } ax^2+bx+c \text{ is } 4(b^2-ac) < 0$$

$\therefore$ the equation has imaginary roots

Linked comprehension type questions

Passage - I:

24-26. $(1+ax+bx^2+cx^3)^{10}$

$$= 1 + p_1x + p_2x^2 + p_3x^3 + \dots + p_{30}x^{30}$$

$$10(a+2bx+3cx^2)(1+ax+bx^2+cx^3)^9$$

$$= p_1 + 2p_2x + 3p_3x^2 + \dots + 30p_{30}x^{29}$$

$$\Rightarrow 10(a+2bx+3cx^2)(1+p_1x+p_2x^2+\dots+p_{30}x^{30})$$

$$= (1+ax+bx^2+cx^3)(p_1+2p_2x^2+3p_3x^3+\dots+30p_{30}x^{29})$$

Equation g like terms on both sides, we get the required values

Passage - II:

$$27. \sum_{0 \leq i < j \leq 20} \sum_{i=0}^{20} \sum_{j=0}^{20} a_i a_j = \frac{\sum_{i=0}^{20} \sum_{j=0}^{20} a_i a_j - \sum_{i=0}^{20} a_i^2}{2}$$

$$= \frac{1}{2} (2^{40} - 40 C_{20}) = 2^{39} - 39 C_{19}$$

$$28. S = \sum_{0 \leq i < j \leq 20} \sum_{i=0}^{20} \sum_{j=0}^{20} (a_i - a_j)^2$$

$$\Rightarrow 2S = \sum_{i=0}^{20} \sum_{j=0}^{20} (a_i - a_j)^2 = \sum_{i=0}^{20} \sum_{j=0}^{20} (a_i^2 + a_j^2 - 2a_i a_j)$$

$$2S = 21 \times 2 \times 2^{40} C_{20} - 2 \cdot 2^{40} \Rightarrow S = 42^{39} C_{19} - 2^{40}$$

$$29. S = \sum_{0 \leq i < j \leq 20} \sum_{i=0}^{20} \sum_{j=0}^{20} (i+j) a_i a_j$$

$$= \sum_{i \leq j \leq 20} \sum_{i=0}^{20} \sum_{j=0}^{20} (20-i+20-j) a_i a_j$$

$$2S = 40 \sum_{0 \leq i < j \leq 20} \sum_{i=0}^{20} \sum_{j=0}^{20} a_i a_j \Rightarrow S = 20(2^{39} - 39 C_{19})$$

Matrix matching type questions

30. A) $(2n_{C_0})^2 - (2n_{C_1})^2 + (2n_{C_2})^2 - \dots + (2n_{C_n})^2 = 2n_{C_n}$
 B) $(2n_{C_0})^2 + (2n_{C_1})^2 + (2n_{C_2})^2 - \dots + (2n_{C_n})^2 = 4n_{C_n}$
 C) $C_1^2 + 2C_2^2 + 3C_3^2 + \dots + nC_n^2 = n(2n-1)C_{n-1}$
 D) $(2n+1)C_n^n - (2n+1)C_n^{n-1} + (2n+1)C_n^{n-2} - \dots = 0$

$$31. C = \sum_{r=0}^{100} (-1)^r (100C_r)^2 = 100C_{50}$$

$$B = \sum_{r=0}^{50} (50C_r)^2 = 100C_{50}$$

$$A = 100C_{50} - 1$$

Integer answer type questions

$$32. C_1 - C_2 \left(1 + \frac{1}{2}\right) + C_3 \left(1 + \frac{1}{2} + \frac{1}{3}\right) - \dots$$

$$+ (-1)^{n-1} C_n \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right)$$

$$\int_0^1 (C_1 - C_2(1+n) + C_3(1+n+n^2) - \dots$$

$$+ (-1)^{n-1} C_n (1+x+x^2 + \dots + x^{n-1})) dx$$

$$= \int_0^1 \frac{1}{1-x} (C_1 - C_2 + C_3 - C_4 + \dots + (-1)^{n-1} C_n)$$

$$- (C_1 x + C_2 x^2 + C_3 x^3 - \dots + (-1)^{n-1} (n x^n)) dx$$

$$\int_0^1 \frac{(C_0 - C_1 x + C_2 x^2 - \dots + (-1)^n C_n x^n)}{1-x} dx$$

$$= \int_0^1 (1-x)^{n-1} dx = \frac{1}{n} \text{ Here, } n = 2015$$

$$33. a = \frac{1}{3^{323}} + 1 \Rightarrow f(x) = {}^nC_0 a^{n-1} - {}^nC_1 a^{n-2} + {}^nC_2 a^{n-3} - \dots$$

$$+ (7)^{n-1} {}^nC_{n-1} a^0 + (-1)^n (-1)^n$$

$$f(n) = \frac{(a-1)^n - (-1)^n}{a} \Rightarrow f(2007) + f(2008) = 3^K$$

$$= \frac{(a-1)^{2007} - (-1)^{2007} + (a-1)^{2008} - (-1)^{2008}}{a}$$

$$= \frac{(a-1)^{2007} + (a-1)^{2008}}{a} \Rightarrow K=9$$

$$34. {}^{2n}C_n = \text{Coeff of } x^n \text{ in } (1+x)^{2n} : {}^{2n-1}C_n$$

$$= \text{Coeff of } x^n \text{ in } (1+x)^{2n-1}$$

$$\text{LHS: Coeff of } x^n \text{ in } [C_0(1+x)^{2n} - C_1(1+x)^{2n-1} + \dots$$

$$+ (-1)^n C_n(1+x)^0]$$

$$\text{coeff of } x^n \text{ in } (1+x)^n [(1+x) - 1]^n \Rightarrow \text{Coeff of } x^n \text{ is } 1$$

35. If S be the sum of coefficients in the expansion of

$$(px + qy - rz)^n \text{ (p, q, r > 0), then } \lim_{n \rightarrow \infty} \frac{S}{(S^{1/n} + 1)^n}$$

$$\text{Ans } \lim_{n \rightarrow \infty} \frac{(p+q-r)^{1/2}}{p+q-r+1} = \frac{0}{p+q-r+1} = 0$$

$$36. \lim_{n \rightarrow \infty} C_n - C_{n-1} \left(\frac{2}{3}\right) + C_{n-2} \left(\frac{2}{3}\right)^2 - \dots + (-1)^n {}^nC_0 \left(\frac{2}{3}\right)^n$$

$$\Rightarrow \left(1 - \frac{2}{3}\right)^n = \lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n = 0$$

$$37. (1+3x+2x^2)^6 \Rightarrow (x+1)^6(2x+1)^6$$

$$\Rightarrow \text{Coeff of } x^{11} = {}^6C_6 \times {}^6C_5 \cdot 2^5 + {}^6C_5 \cdot {}^6C_6 \cdot 2^6 = 2^6 \times 3^2$$

$$38. f(m,n) = {}^nC_0 - {}^nC_1 + {}^nC_2 - \dots + (-1)^{m-1} {}^nC_{m-1} \Rightarrow f(7,8)$$

$$= {}^8C_0 - {}^8C_1 + {}^8C_2 - {}^8C_3 + {}^8C_4 - {}^8C_5 + {}^8C_6 - {}^8C_7 = 7$$

EXERCISE-III

LEVEL-I (MAIN)

Single Answer type questions

$$1. \left| \frac{4x}{3} \right| < 1$$

$$2. \left| \frac{2x}{a} \right| < 1$$

$$3. 1 < x < \frac{3}{2} \Rightarrow (3+2x)^{3/5} = (2x)^{3/5} \left(1 + \frac{3}{2x}\right)^{3/5}$$

$$T_3 = (2x)^{3/5} (-3) \left(\frac{3}{10x}\right)^2$$

4. $|2x| < 1$ and $|3x| < 1$

5. $G.E = (1-x)^{-2}$, $|x| < 1$

6. $T_{r+1} = \frac{n(n-1)\dots(n-r+1)}{r!} (-x)^r, n = \frac{1}{2}$

7. $(2a-3b)^{-\frac{1}{2}} = (2a)^{-\frac{1}{2}} \left(1 - \frac{3b}{2a}\right)^{-\frac{1}{2}}$ and write general term

8. $(1-x)^{-2}$, coefficient of $x^n = x+1$

9. $(1-x)^{-2}$, coefficient of $x^{24} = 25$

10. $1 + (1+2)x + (1+2+3)x^2 + \dots = (1-x)^{-2}$

11. $(1-x)^{-4} = 1 + 4x + 10x^2 + \dots \infty$

12. $(1+2x+x^2) 8^{-1/3} \left(1 - \frac{x}{8}\right)^{-1/3}$
 $= \frac{1}{2} (1+2x+x^2) \left(1 + \frac{x}{24} + \frac{x^2}{288} + \dots\right)$

13. Coefficient of x^n in $\frac{(1+x)^{2n}}{x^n}$ is 1

14. $(1+2x)(1-2x)^{-2}$, coefficient of x^r is $(r+1)2^r + r2^r$

15. $\frac{1}{(1-ax)(1-bx)} = \left(\frac{a}{a-b}\right)(1-ax)^{-1} + \left(\frac{b}{b-a}\right)(1-bx)^{-1}$

coefficient of x^n is $\frac{a \cdot a^n}{a-b} + \frac{b \cdot b^n}{b-a}$

16. $(1+x^{12}+x^{24}+x^{36}) (1+^{12}C_1 x^2 + ^{12}C_2 x^4 + \dots)$

coefficient of x^{24} is $^{12}C_{12} + ^{12}C_6 + 1$

17. T_8

18. $2 + \frac{35x}{6}$

19. $\left(1 + \frac{c}{l}\right)^{\frac{-1}{2}} + \left(1 - \frac{c}{l}\right)^{\frac{-1}{2}}$
 $= \left(1 - \frac{1}{2} \frac{c}{l} + \frac{3}{4} \frac{c^2}{l^2} \dots\right) + \left(1 + \frac{c}{2l} + \frac{3}{4} \frac{c^2}{l^2} + \dots\right)$

20. $\frac{1}{2} + \frac{31}{64}x$

21. $\left(\frac{p}{q}\right)^K = \frac{n(p+q) + (p-q)}{n(p+q) - (p-q)} = \frac{1 + \left(\frac{p-q}{p+q}\right) \frac{1}{n}}{1 - \left(\frac{p-q}{p+q}\right) \frac{1}{n}}$
 $= \frac{\left(1 + \frac{p-q}{p+q}\right)^{\frac{1}{n}}}{\left(1 - \frac{p-q}{p+q}\right)^{\frac{1}{n}}} = \left(\frac{p}{q}\right)^{1/n}$

22. Let $x = 1+h$ (h is very small)

23. Let $x = 1+h$

24. $S = \frac{1.3}{4.8} - \frac{1.3.5}{4.8.12} + \dots$
 $S = \left(1 - \frac{1}{4} + \frac{1.3}{2!} \left(\frac{1}{4}\right)^2 \dots\right) - \frac{3}{4}$
 $= \left(1 + \frac{1}{2}\right)^{-1/2} - \frac{3}{4} = \sqrt{\frac{2}{3}} - \frac{3}{4}$

25. $S = 3 \left(\frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots\right)$
 $= 3 \left\{1 + \frac{1}{3} + \frac{1.3}{3.6} + \dots\right\} - 4 = 3\sqrt{3} - 4$

26. $S = \frac{1}{2} \left(\frac{2.5}{9.18} + \frac{2.5.8}{9.18.27} + \dots\right)$
 $= \frac{1}{2} \left(1 + \frac{2}{9} + \frac{2.5}{9.18} + \dots\right) - \frac{11}{18} = \frac{1}{2} \left(\left(\frac{9}{4}\right)^{1/3} - \frac{11}{18}\right)$

27. $1 + \frac{n}{2} + \frac{n(n-1)}{2.4} + \dots = \left(\frac{3}{2}\right)^n$

$1 + \frac{n}{3} + \frac{n(n+1)}{3.6} + \dots = \left(\frac{3}{2}\right)^n$

Numerical value type questions

28. $17^{256} = (290-1)^{128} = 1000k + ^{128}C_{126} (290)^2 - ^{128}C_{127} (290) + 1$

$\therefore a_1 = 5, a_2 = 6, a_3 = 1$

29. $(1+x+x^2+x^3+x^4)^{1001} (1-x)^{1002}$
 $= (1-x)(1-x^5)^{1001}$, so all the powers of x will be of $5m$ or $5m+1$ ($m \in I$)

So, coeff of x^{2009} will be 0

30. $(1-2x+5x^2+10x^3) [C_0 + C_1 X + C_2 X^2 + \dots]$
 $= 1 + a_1 x + a_2 x^2 + \dots \Rightarrow a_1 = n-2$ and

$a_2 = \frac{n(n-1)}{2} - 2n + 5$

put $a_1^2 = 2a_2$; $(n-2)^2 = n(n-1) - 4n + 10$

$n^2 - 4n + 4 = n^2 - 5n + 10$; $n = 6$

31. We have $\frac{{}^nC_k}{{}^nC_{k-1}} = \frac{n-k+1}{k}$

$\therefore \sum_{k=1}^n k^3 \left(\frac{{}^nC_k}{{}^nC_{k-1}}\right)^2 = \sum_{k=1}^n k^3 \left(\frac{n-k+1}{k}\right)^2$

$$\begin{aligned}
 &= \sum_{k=1}^n k\{(n+1)-k\}^2 = \sum_{k=1}^n k\{(n+1)^2 - 2(n+1)k + k^2\} \\
 &= \sum_{k=1}^n k\{(n+1)^2 - 2(n+1)k + k^2\} \\
 &= (n+1)^2 \left(\sum_{k=1}^n k \right) - 2(n+1) \left(\sum_{k=1}^n k^2 \right) + \left(\sum_{k=1}^n k^3 \right) \\
 &= (n+1)^2 \left\{ \frac{n(n+1)}{2} \right\} - 2(n+1) \frac{n(n+1)(2n+1)}{6} + \left\{ \frac{n(n+1)}{2} \right\}^2 \\
 &= \frac{n(n+1)^2}{2} \left\{ (n+1) - \frac{2}{3}(2n+1) + \frac{n}{2} \right\} \dots (i) \\
 &= \frac{n(n+1)^2}{2} \times \frac{6(n+1) - 4(2n+1) + 3n}{6} \\
 &= \frac{n(n+1)^2(n+2)}{12}
 \end{aligned}$$

PRACTICE SHEET

EXERCISE-I

LEVEL-I (MAIN)

Single Answer type questions

- $(1-x)(1-x^5)^{1001}$; coefficient of x^{2009} is 0
- $\frac{{}^nC_1 a}{{}^nC_2 b} = \frac{{}^{(n+3)}C_2 a}{{}^{(n+3)}C_3 b} \Rightarrow n = 5$
- $(18+7)^3 / (3+2)^6$
- $(1+0.2)^4 + (1-0.2)^4$
- Coefficient of x , ${}^5C_3 A^3 = 270 \Rightarrow A = 3$
- $r = \frac{np-k}{p+q} = 2$; so coefficient of x^4 is ${}^{10}C_2 (-1)^2 \frac{3^2}{2^8}$
- $r = \frac{np}{p+q} = 5 \Rightarrow$ Required term is T_6
- $\frac{(1+x)^{31}}{x} - \frac{(1+x)^{21}}{x}$,
coefficient of x^5 is ${}^{31}C_6 - {}^{21}C_6$
- ${}^9C_2 3^7 a^2 = {}^9C_3 3^6 a^3 \Rightarrow a = 9/7$
- ${}^{20}C_n = {}^{20}C_{n+2} \Rightarrow n = 9$
- ${}^{2n}C_{3n-1} = {}^{2n}C_{r+1}$

- $T_6 = {}^{10}C_5 \left(\frac{2x^2}{3} \right)^5 \left(\frac{3}{x^2} \right)^5 = 252$
- T_{n+1} in the given expansion
- $\left(x + \frac{1}{x} \right)^{2n}$, middle term in $T_{n+1} = {}^{2n}C_n$
- For $n \in N$, two consecutive terms in the expansion of $(p+q)^n$ are equal
 $\Rightarrow \frac{(n+1)q}{p+q}$ is a positive integer.
- $\frac{(n+1)(y)}{|x|+|y|} = 18.666$
- ${}^nC_{r-1} = 45, {}^nC_r = 120, {}^nC_{r+1} = 210$
 $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{120}{45} \Rightarrow \frac{n-r+1}{r} = \frac{8}{3}$
 $\frac{{}^nC_{r+1}}{{}^nC_r} = \frac{210}{120} \Rightarrow \frac{n-r}{r+1} = \frac{7}{4} \Rightarrow n = 10$
- $T_{r+1} = {}^{10}C_r 2^{\frac{10-r}{2}} 3^{\frac{r}{5}}$ For term to be rational,
 $\frac{10-r}{2} \in I$ and $\frac{r}{5} \in I \Rightarrow r = 0, 10 \therefore t_1 + t_{11} = 41$
- $T_{r+1} = {}^{256}C_r 3^{\frac{256-r}{2}} 5^{\frac{r}{8}}$ For term to be rational,
 $\frac{256-r}{2} \in I, \frac{r}{8} \in I$
- $T_{r+1} = {}^{24}C_r 3^{\frac{24-r}{5}} 2^{\frac{r}{7}}$, where $r = 14$
- No. of terms in $\left(x - \frac{1}{x} \right)^n$ is $2n+1$
- $\sum \frac{18!}{n_1! n_2! n_3!} a^{n_1} b^{n_2} c^{n_3}$ where $n_1 + n_2 + n_3 = 18$
- $-(1+4+9+\dots+100)$
- $1+4+9+100$
- $(1+15)^n - 15n - 1$ is divisible by 15^2
- $(2n-1)^n + (2n)^n < (2n+1)^n, n > 2$ and $n \in N$
- $\frac{3^{1001}}{82} = 3 \cdot \frac{(3^4)^{250}}{82} = \frac{3}{82} (82-1)^{250} \cdot \left\{ \frac{3^{1001}}{82} \right\} = \frac{3}{82}$